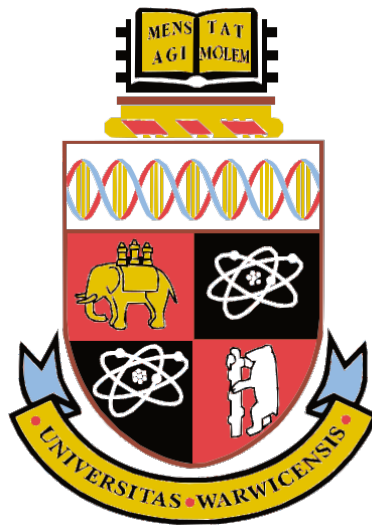




The Symbolic Method: From Classical Invariant Theory to Modern Geometric Invariant Theory

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Contents

1	The Strange Case of Professor Hilbert and Invariant Theory	1
2	Symbols, Brackets, and all that Jazz	2
2.1	Back to the basics	2
2.2	Polarisation and the Cayley Ω -Operator	4
2.3	Symbolic Method as a Computational Tool	8
2.4	The First Fundamental Theorem	10
2.5	Multilinearity of Symbolic Brackets on Sums of Powers	11
3	The Old, the New, and the True	13
3.1	Binary Forms and the $SL_2(\mathbb{C})$ -Action	13
3.2	The Discriminant in Low Degree	14
3.3	The Hessian Covariant for Binary Cubics	15
3.4	Transvectants	17
3.5	The General Algorithm for Invariants	19
3.6	A Worked Example: The Invariant I of the Quartic	20
3.7	Lifting to Dimension 3	21
3.8	Plane Cubics, the Hesse Pencil, and the Geometry of S and T	22
4	The Bigger Picture	23
4.1	Geometric Loci of Quartic Invariants	23
4.2	Geometric Loci of Ternary Cubics via the Hesse Pencil	25
4.3	Generic Rank and Secant Varieties	27
5	A Brief Expedition into Geometric Invariant Theory	29
5.1	The Quotient $\mathbb{P}(V_d)SL_2$ as a Moduli Space	29
5.2	Plane Curves and their GIT Quotients	31
5.3	Covariants and the Rationality of Moduli Plane Curves	33
6	The Noether Code	34
6.1	Implementing Symbolic Contraction via the Cayley- Ω Operator	35
6.2	Geometric Evaluation on the Hesse Pencil	35
6.3	Searching for Invariants	36
6.4	Example Output	37
7	The Classics Do Indeed Live On	38
A	Source Code: The Noether Code	39
A.1	Module 1: noether_engine.py	39
A.2	Module 2: evaluator.py	43
A.3	Module 3: groebner_checker.py	44
A.4	Module 4: main.py	45
	Bibliography	48

1 The Strange Case of Professor Hilbert and Invariant Theory

David Hilbert was a mathematician of such formidable intellect that he managed to single-handedly almost kill invariant theory [Par90]. The attempt came about fifty years after George Boole's 1842 paper, *Exposition of a General Theory of Linear Transformations* [Boo42]. Boole's work fuelled the beginnings of algebraic invariant theory, a branch of study that would receive contributions from numerous mathematicians, namely Arthur Cayley, J.J. Sylvester, Alfred Clebsch, and Paul Gordan [Cri88]. Initially, the English empirical school, led by Cayley and Sylvester, approached the classification of polynomial forms constructively, relying on extensive symbolic computation and differential operators to explicitly calculate invariants [Olv06; Par89].

As the degrees and dimensions of these forms grew, direct computation quickly became unmanageable. To bypass this, the German school, led by Clebsch and Gordan, developed the *symbolic method* [Cri88]. By representing forms as formal powers of symbolic linear factors, they constructed invariants purely through antisymmetric bracket pairings. This symbolic calculus revealed itself in Gordan's 1868 finiteness theorem, which constructively proved that every binary form possesses a finite generating basis of invariants [GY03; Stu08]. However, in 1890, Hilbert generalized Gordan's finiteness result to forms in any number of variables using an abstract, non-constructive proof by contradiction [Hil93]. While Hilbert solved the central question of the field, his lack of an algorithmic method for actually finding those invariants marked a significant shift in the direction of the field.. Explicit algorithmic approaches was mostly abandoned in favour of abstract commutative algebra, leaving the classical symbolic method dormant for nearly a century [Par90].

The symbolic method resurrected in the late twentieth century by Gian-Carlo Rota and his collaborators [DRS74; RS86]. Rather than treating the method as a motivating argument, Rota proved that it is deeply grounded in combinatorial foundations and the representation theory of the symmetric group [Gro03; Rot01]. The revival has contributed to a modern resurgence of computational invariant theory, producing advanced algorithms for calculating invariant rings in modern computer algebra systems [DK15; DRS25].

Simultaneously, the classical theory was combined into the modern framework of Geometric Invariant Theory (GIT), founded by David Mumford [MFK94]. Where nineteenth-century mathematicians sought explicit polynomials, GIT uses these invariants to construct quotient varieties and moduli spaces [Neu07]. Explicit symbolic computations exactly cut out geometric loci, directly linking classical tensor contractions to modern algebraic geometry [Muk03; Dol12]. This intersection remains an active area of research, with applications spanning the rationality of moduli spaces of plane curves [BB08; BBK09], the Waring problem and symmetric tensor rank [Com+08; CCG12], and the detailed geometry of the Hesse pencil for genus one curves [AD09; Fis19; Roc25].

This dissertation aims to connect the historical, algorithmic, and geometric aspects of the symbolic method, using it as a unifying framework for understanding the invariants and covariants of algebraic forms. We begin by formally establishing the symbolic method before extending the symbolic calculus to quantic forms via the Cayley operator. We then compute explicit brackets to study the symmetric tensor rank and the geometric loci of secant varieties. Finally, we apply these symbolic constructions to the moduli space of elliptic curves and the Hesse pencil, demonstrating how classical algebraic machinery explicitly encodes the modern geometric structure of SL_n -quotient varieties.

2 Symbols, Brackets, and all that Jazz

While symbolic notation can first appear opaque, it provides a remarkably efficient language for expressing deep algebraic structures. As discussed in the introduction, the symbolic method provides a systematic framework for constructing and representing invariants and covariants of homogeneous polynomials. We begin by establishing this method and walk through a couple of first-examples.

2.1 Back to the basics

Throughout this section, $V \cong \mathbb{C}^n$.

A *symbol* is defined to be a formal n -tuple of variables

$$a \doteq (\alpha_1, \alpha_2, \dots, \alpha_n),$$

which we interpret as the coefficients of a linear form

$$(ax) \doteq \alpha_1 x_1 + \alpha_2 x_2 + \dots + \alpha_n x_n.$$

Given symbols $a^{(1)}, a^{(2)}, \dots, a^{(n)}$, where each

$$a^{(i)} = (\alpha_1^{(i)}, \alpha_2^{(i)}, \dots, \alpha_n^{(i)})$$

, we define their *bracket* as the $n \times n$ determinant

$$[a^{(1)} a^{(2)} \dots a^{(n)}] \doteq \det \begin{pmatrix} \alpha_1^{(1)} & \alpha_2^{(1)} & \dots & \alpha_n^{(1)} \\ \alpha_1^{(2)} & \alpha_2^{(2)} & \dots & \alpha_n^{(2)} \\ \vdots & \vdots & \ddots & \vdots \\ \alpha_1^{(n)} & \alpha_2^{(n)} & \dots & \alpha_n^{(n)} \end{pmatrix}.$$

Lemma 2.1 (Invariance of the Bracket). *Let $V \cong \mathbb{C}^n$ be the standard representation of $SL_n(\mathbb{C})$,*

a left group action. Let $V^\vee$ be the dual space, equipped with the induced left action:

$$g \cdot a = a g^{-1}$$

for any $g \in SL_n(\mathbb{C})$ and any symbol $a = (\alpha_1, \dots, \alpha_n) \in V^\vee$ viewed as a row vector in $V^\vee$.

Given n symbols $a^{(1)}, a^{(2)}, \dots, a^{(n)}$, their bracket is defined as the $n \times n$ determinant:

$$[a^{(1)} a^{(2)} \dots a^{(n)}] \doteq \det \begin{pmatrix} \alpha_1^{(1)} & \alpha_2^{(1)} & \dots & \alpha_n^{(1)} \\ \alpha_1^{(2)} & \alpha_2^{(2)} & \dots & \alpha_n^{(2)} \\ \vdots & \vdots & \ddots & \vdots \\ \alpha_1^{(n)} & \alpha_2^{(n)} & \dots & \alpha_n^{(n)} \end{pmatrix}.$$

Then the bracket is SL_n -invariant. That is,

$$[g \cdot a^{(1)}, g \cdot a^{(2)}, \dots, g \cdot a^{(n)}] = [a^{(1)} a^{(2)} \dots a^{(n)}].$$

Proof. Let A be the $n \times n$ matrix whose i -th row is the symbol $a^{(i)}$. By definition, the bracket is the determinant of A :

$$[a^{(1)} \dots a^{(n)}] = \det(A).$$

Under the left action, each symbol transforms as $g \cdot a^{(i)} = a^{(i)} g^{-1}$. Because matrix multiplication is performed row by row, transforming all symbols simultaneously corresponds to multiplying the matrix A on the right by g^{-1} .

Hence, the transformed bracket is given by the determinant of the transformed matrix:

$$[g \cdot a^{(1)}, g \cdot a^{(2)}, \dots, g \cdot a^{(n)}] = \det(A g^{-1}).$$

By the multiplicativity of the determinant, this factors as:

$$\det(A g^{-1}) = \det(A) \det(g^{-1}).$$

Since $g \in SL_n(\mathbb{C})$, we have $\det(g) = 1$, which implies $\det(g^{-1}) = 1$. Therefore,

$$[g \cdot a^{(1)}, g \cdot a^{(2)}, \dots, g \cdot a^{(n)}] = \det(A) \cdot 1 = [a^{(1)} a^{(2)} \dots a^{(n)}].$$

This proves that the bracket is an SL_n -invariant. □

Thus the bracket is an alternating SL_n -invariant multilinear form, unique up to a scalar, on the n -dimensional space of symbols.

For the $n = 2$ case, given a symbol $a = (\alpha_1, \alpha_2)$, then we have $(ax) = \alpha_1 x + \alpha_2 y$, which is a

linear form in two variables $(x, y)^t$. By the previous lemma, under $g \in SL_2(\mathbb{C})$ we have

$$(ax) \mapsto (ag^{-1})(x, y)^t = (aX)$$

$$(gx) = a(g^{-1}g)x = ax = (ax).$$

Because the inner product matrices cancel perfectly, the symbolic linear form (ax) is preserved under the combined action, and therefore defines an SL_2 -equivariant pairing. This ensures that arbitrary polynomial expressions built from linear factors (ax) and brackets $[a^{(1)}a^{(2)}]$ behave naturally under SL_2 .

2.2 Polarisation and the Cayley Ω -Operator

We now introduce an additional space. Maintaining $V \cong \mathbb{C}^n$, we denote by

$$V_{n,d} = \text{Sym}^d(V^\vee)$$

the space of homogeneous quantic forms of degree d .

The symbolic method represents a homogeneous polynomial as a symmetrised product of linear forms. Geometrically, this formalism corresponds to viewing $V_{n,d}$ as embedded in the tensor product of d copies of $V^\vee$. By treating the form as if it were a pure d -th power of a single linear form, we can study invariants of $V_{n,d}$ by looking at the permutation-invariant elements of this larger tensor space.

To pass from a general polynomial to such a representation, we use the process of *polarisation*. Polarisation associates to a homogeneous polynomial f of degree d a symmetric multilinear form $\tilde{f}$ in d vector variables. When all d vectors are equal to x , we recover the original polynomial: $\tilde{f}(x, \dots, x) = f(x)$. This reduces the study of polynomial invariants to multilinear ones, which can then be algorithmically computed using determinant operations and differential operators like the Cayley Ω -process.

A *quantic form* of degree d is a homogeneous polynomial in n variables $x = (x_1, \dots, x_n)$, written as

$$f(x) = \sum_{|I|=d} \binom{d}{I} a_I x^I \in V_{n,d},$$

where $I = (i_1, \dots, i_n)$ is a multi-index of non-negative integers such that $|I| = i_1 + \dots + i_n = d$. Here, x^I denotes the monomial $x_1^{i_1} \dots x_n^{i_n}$, and the multinomial coefficient is given by

$$\binom{d}{I} = \frac{d!}{i_1! \dots i_n!}.$$

The sequence of coefficients $(a_I)_{|I|=d}$ provides a system of linear coordinates on the vector

space $V_{n,d} \cong \mathbb{C}^N$, where

$$N = \binom{n+d-1}{d}$$

is the number of distinct monomials of degree d in n variables.

A *symbolic representation* of $f \in V_{n,d}$ is an expression of the form

$$f(x) = \Phi(a^{(1)}, \dots, a^{(d)}) (a^{(1)}x) \cdots (a^{(d)}x),$$

where $a^{(1)}, \dots, a^{(d)}$ are symbols and Φ is a polynomial invariant under permutations of the $a^{(i)}$, chosen so that expanding and identifying coefficients recovers f . Compactly, Φ is symmetric in the symbols.

In practice, for a generic form, we use a single symbol a and write

$$f(x) = (ax)^d,$$

with the understanding that the coefficients are identified after expansion,

$$(ax)^d = \sum_{|I|=d} \binom{d}{I} a^I x^I = \sum_{i_1+\dots+i_n=d} \frac{d!}{i_1! \cdots i_n!} \alpha_1^{i_1} \cdots \alpha_n^{i_n} x_1^{i_1} \cdots x_n^{i_n}$$

with the coefficient pattern of f . This representation is justified by the fact that the map

$$(\alpha_1, \dots, \alpha_n) \mapsto (ax)^d$$

realises $V_{n,d}$ as the image of the d -th Veronese embedding of $\mathbb{P}^{n-1}$. Since the d -th powers of linear forms span $V_{n,d}$, the symbolic parameters $(\alpha_1, \dots, \alpha_n)$ parametrise a dense open subset of $V_{n,d}$. Geometrically, the powers $(ax)^d$ form the image of the d -th Veronese embedding of $\mathbb{P}^{n-1}$.

For example, consider a quadratic $f(x, y) = ax^2 + 2bxy + cy^2$ we write

$$f(x, y) = (ax)^2 = \alpha_1^2 x^2 + 2\alpha_1 \alpha_2 xy + \alpha_2^2 y^2,$$

and identify $a = \alpha_1^2$, $2b = 2\alpha_1 \alpha_2$, $c = \alpha_2^2$ as before. The discriminant

$$\Delta = ac - b^2$$

is represented symbolically as,

$$\Delta = (\alpha_1^2 \alpha_2^2) - (\alpha_1 \alpha_2)^2 = 0,$$

which vanished identically if only a single symbol is used. This reflects the fact that invariants require at least two independent symbolic copies.

Let $a^{(1)}, a^{(2)}, \dots$ be symbols. Consider monomials built from:

- factors of the form $(a^{(i)}x)$;
- quantic bracket factors $[a^{(j_1)} \dots a^{(j_n)}]$.

Each symbol contributes two quantities:

- 1 to the *order* for each time it appears linearly as $(a^{(i)}x)$;
- 1 to the *coefficient degree* for each time it appears in any factor (either linear or bracket).

A *bracket monomial* an expression of the form

$$M = \prod_i (a^{(i)}x)^{u_i} \prod_J [a^{(j_1)} \dots a^{(j_n)}]^{v_J}$$

such that each $a^{(i)}$ is a symbol, $J = (j_1, \dots, j_n)$ indexes n -tuples of symbols, the exponents u_i, v_J are non-negative integers, with only finitely many non-zero exponents.

Suppose each symbol $a^{(i)}$ is substituted by a linear form whose coefficients are those of some quantic form. Expanding in the symbols, we obtain a polynomial in the coefficients of the chosen forms and in $x = (x_1, \dots, x_n)$.

For this chapter we restrict to the case of *one* underlying form $f \in V_{n,d}$, represented symbolically as $(a^{(1)}x)^d$. Any other symbol $a^{(i)}$ will be taken to represent the same form, but with independent formal coefficients; the identification $a^{(1)} \sim a^{(2)}$ is enforced only after expansion.

Proposition 2.2. *Let $f(x) = (ax)^d \in V_{n,d}$ and let M be a bracket monomial built from symbols $a^{(1)}, a^{(2)}, \dots$ as above. After expanding M and identifying all symbols with a single symbol a , the resulting polynomial is $SL_n(\mathbb{C})$ -equivariant.*

Proof. The group $SL_n(\mathbb{C})$ acts on V from the left, and on $V^\vee$ via the left action $g \cdot \alpha = \alpha g^{-1}$.

A bracket monomial M involving r distinct symbols $a^{(1)}, \dots, a^{(r)}$ is a formal product of linear factors and bracket factors. Geometrically, we view each symbol $a^{(i)}$ as a vector in $V^\vee$. The brackets evaluate the unique $SL_n(\mathbb{C})$ -invariant alternating multilinear volume form $\omega \in \Lambda^n V$. The linear factors $(a^{(i)}x)$ correspond to the canonical pairing $V^\vee \otimes V \rightarrow \mathbb{C}$.

Because M is constructed entirely from complete contractions of tensors using the invariant form ω and the natural pairing, the multilinear map:

$$\tilde{M} : \underbrace{V^\vee \otimes \dots \otimes V^\vee}_{r \text{ times}} \times V \rightarrow \mathbb{C}$$

defined by the bracket monomial is explicitly $SL_n(\mathbb{C})$ -equivariant.

The process of “identifying” the symbols $a^{(1)} = \dots = a^{(r)} = a$ is equivalent to restricting this map to the diagonal of the product, which factors through the map:

$$\pi : \text{Sym}^r(V^\vee) \rightarrow V^\vee \otimes \dots \otimes V^\vee.$$

Since $SL_n(\mathbb{C})$ acts diagonally on tensor products, the map π is an $SL_n(\mathbb{C})$ -morphism. Therefore, the composition of substituting a specific quantic form $f \in V_{n,d}$ into the equivariant tensor $\tilde{M}$ commutes with the action of $SL_n(\mathbb{C})$. Hence, the resulting polynomial transforms compatibly with the group action. This follows from the fact that all contractions are performed using SL_n -invariant tensors. $\square$

The main task of the symbolic method is therefore combinatorial. We construct bracket monomials with the prescribed order and coefficient degree.

Remark 2.3. If no factor of the form $(a^{(i)}x)$ appears, the resulting expression is independent of x . Thus, invariants correspond to monomials consisting purely of brackets.

The *Cayley Ω -operator* associated to variables $x^{(1)}, \dots, x^{(n)}$ is defined by the determinant of the matrix of partial derivatives:

$$\Omega \doteq \det \begin{pmatrix} \frac{\partial}{\partial x_1^{(1)}} & \cdots & \frac{\partial}{\partial x_n^{(1)}} \\ \vdots & \ddots & \vdots \\ \frac{\partial}{\partial x_1^{(n)}} & \cdots & \frac{\partial}{\partial x_n^{(n)}} \end{pmatrix} = \sum_{\sigma \in S_n} \text{sgn}(\sigma) \frac{\partial}{\partial x_{\sigma(1)}^{(1)}} \cdots \frac{\partial}{\partial x_{\sigma(n)}^{(n)}}.$$

A fundamental identity is that Ω behaves like contraction with the bracket. In particular, we find the following formula.

Proposition 2.4. *Let $f_1, \dots, f_n$ be quantic forms of degrees $d_1, \dots, d_n$. If we represent them symbolically as $f_i(x^{(i)}) = (a^{(i)}x^{(i)})^{d_i}$, then for each integer $r \geq 0$ the following identity holds:*

$$\Omega^r (f_1(x^{(1)}) \cdots f_n(x^{(n)})) = \left(\prod_{i=1}^n \frac{d_i!}{(d_i - r)!} \right) [a^{(1)} \dots a^{(n)}]^r \prod_{i=1}^n (a^{(i)}x^{(i)})^{d_i - r}.$$

Proof. Let $f_i(x^{(i)}) = (a^{(i)}x^{(i)})^{d_i}$ be the symbolic representations of the quantic forms, where $(a^{(i)}x^{(i)}) = \alpha_1^{(i)}x_1^{(i)} + \dots + \alpha_n^{(i)}x_n^{(i)}$.

We compute the action of Ω on the symbolic product. Using the chain rule, differentiating f_i with respect to its coordinates brings down the corresponding symbolic coefficients. Specifically, expanding a single application of Ω gives:

$$\Omega \left(\prod_{i=1}^n (a^{(i)}x^{(i)})^{d_i} \right) = \sum_{\sigma \in S_n} \text{sgn}(\sigma) \prod_{i=1}^n \frac{\partial}{\partial x_{\sigma(i)}^{(i)}} (a^{(i)}x^{(i)})^{d_i}.$$

Because $(a^{(i)}x^{(i)}) = \alpha_1^{(i)}x_1^{(i)} + \dots + \alpha_n^{(i)}x_n^{(i)}$, differentiating with respect to $x_{\sigma(i)}^{(i)}$ brings down an $\alpha_{\sigma(i)}^{(i)}$ (in direct analogy with ordinary differentiation). Specifically:

$$\frac{\partial}{\partial x_{\sigma(i)}^{(i)}} (a^{(i)}x^{(i)})^{d_i} = d_i \alpha_{\sigma(i)}^{(i)} (a^{(i)}x^{(i)})^{d_i-1}.$$

However, notice that the symbolic coefficients inside the sum exactly form the expansion of the bracket determinant:

$$\sum_{\sigma \in S_n} \text{sgn}(\sigma) \alpha_{\sigma(1)}^{(1)} \dots \alpha_{\sigma(n)}^{(n)} = \det \begin{pmatrix} \alpha_1^{(1)} & \dots & \alpha_n^{(1)} \\ \vdots & \ddots & \vdots \\ \alpha_1^{(n)} & \dots & \alpha_n^{(n)} \end{pmatrix} = [a^{(1)} \dots a^{(n)}].$$

Thus, applying Ω to the symbolic product effectively multiplies the remaining factors by the bracket $[a^{(1)} \dots a^{(n)}]$, yielding:

$$\Omega \left[\prod_{i=1}^n (a^{(i)}x^{(i)})^{d_i} \right] = \left(\prod_{i=1}^n d_i \right) [a^{(1)} \dots a^{(n)}] \prod_{i=1}^n (a^{(i)}x^{(i)})^{d_i-1}.$$

Iterating the differential operator r times yields the r -th power of the bracket and accumulates the factorial scaling factors. This completes the proof. $\square$

Thus, bracket constructions are realised through repeated application of the differential operator Ω .

Let $f(x) = (ax)^d$ be a quantic form of degree d . The Hessian determinant of f can be computed via the Ω -operator. By introducing n independent copies of f with symbols $a^{(1)}, \dots, a^{(n)}$ and coordinates $x^{(1)}, \dots, x^{(n)}$, we evaluate:

$$\Omega^2(f(x^{(1)}) \dots f(x^{(n)})) \Big|_{x^{(i)}=x} = \left(\frac{d!}{(d-2)!} \right)^n [a^{(1)} \dots a^{(n)}]^2 \prod_{i=1}^n (a^{(i)}x)^{d-2}.$$

After evaluation and identification $a^{(1)} \sim \dots \sim a^{(n)} \sim a$, this yields a scalar multiple of the $n \times n$ Hessian of f . Symbolically, we contracted the factors via the repeated application of Ω using the bracket $[a^{(1)} \dots a^{(n)}]$.

2.3 Symbolic Method as a Computational Tool

Having established the symbolic method, we now examine its use in constructing invariants.

Following the classical approach of Grace and Young [GY03], we define invariants constructively. For a quantic form $f \in V_{n,d}$, an invariant is a polynomial in the coefficients of f

that remains invariant under the induced action when the coordinate variables $x_1, \dots, x_n$ undergo a general linear transformation.

The key computational principle is that invariants are obtained by fully contracting symbolic expressions so that all dependence on x is eliminated.

Suppose that $f(x) = (ax)^d \in V_{n,d}$ and we introduce r independent symbols $a^{(1)}, \dots, a^{(r)}$, all representing the same underlying form f . Consider the completely symbolic product

$$F(x^{(1)}, \dots, x^{(r)}) \doteq (a^{(1)}x^{(1)})^d (a^{(2)}x^{(2)})^d \dots (a^{(r)}x^{(r)})^d.$$

This expression is homogeneous of degree d in each symbol $a^{(i)}$.

By applying a carefully chosen polynomial in the Cayley Ω -operators to F , and finally enforcing the identification $a^{(1)} \sim \dots \sim a^{(r)} \sim a$, we obtain an expression evaluated purely in the coefficients of f alone. Because each application of Ω extracts a quantic bracket, this differential process corresponds exactly to the combinatorial construction of bracket polynomials.

In practice, computations are often done directly at the level of these bracket monomials. We illustrate this in the simplest non-trivial case, a binary quadratic.

Let $f(x, y) = ax^2 + 2bxy + cy^2$ be a binary quadratic form. We introduce two independent symbols $a = (\alpha_1, \alpha_2)$ and $b = (\beta_1, \beta_2)$ representing f . Their linear forms are:

$$(ax) = \alpha_1 x + \alpha_2 y, \quad (bx) = \beta_1 x + \beta_2 y.$$

The fundamental 2×2 bracket is $[ab] = \alpha_1 \beta_2 - \alpha_2 \beta_1$. Because each symbol must appear exactly $d = 2$ times to match the degree of the form, the simplest invariant that can be constructed is the squared bracket:

$$I = [ab]^2 = (\alpha_1 \beta_2 - \alpha_2 \beta_1)^2.$$

This expression is homogeneous of degree 2 in the coefficients and independent of (x, y) .

To evaluate this symbolically, we expand the square:

$$[ab]^2 = \alpha_1^2 \beta_2^2 - 2\alpha_1 \alpha_2 \beta_1 \beta_2 + \alpha_2^2 \beta_1^2.$$

Identifying the symbols yields,

$$\alpha_1^2 \sim \beta_1^2 \sim a, \quad \alpha_1 \alpha_2 \sim \beta_1 \beta_2 \sim b, \quad \alpha_2^2 \sim \beta_2^2 \sim c.$$

Substituting these real coefficients into the expanded bracket yields:

$$[ab]^2 \longmapsto a(c) - 2(b)(b) + c(a) = 2(ac - b^2).$$

Thus, up to a scalar multiple, this recovers the classical discriminant $\Delta = ac - b^2$.

2.4 The First Fundamental Theorem

The previous example demonstrates that the symbolic method is a powerful engine for constructing invariants out of determinant brackets. This raises the natural question: does this procedure generate all invariants?

The answer, commonly known as the First Fundamental Theorem (FFT), was one of the crowning achievements of classical invariant theory by Cayley and Sylvester during the 19th century. It guarantees that the purely algebraic machinery of the symbolic method is not just a convenient tool, but a complete classification of the invariant ring.

Theorem 2.5 (First Fundamental Theorem for SL_n). *The $\mathbb{C}$ -algebra of $SL_n(\mathbb{C})$ -invariant polynomials in finitely many vector variables $x^{(1)}, x^{(2)}, \dots, x^{(n)} \in V$ is generated by the $n \times n$ determinant brackets $[x^{(j_1)} \dots x^{(j_n)}]$.*

Proof. The proof proceeds in three steps: reduction to multilinear polynomials via polarisation, upgrading to absolute $GL_n(\mathbb{C})$ -invariants, and application of the Cayley Ω -process.

Step 1 Polarisation to Multilinear Polynomials: Let $F(x^{(1)}, x^{(2)}, \dots)$ be an $SL_n(\mathbb{C})$ -invariant polynomial, homogeneous of degree d_i in each vector $x^{(i)}$. By the algebraic process of complete polarisation (introducing new independent vectors and applying directional derivatives), we can replace the higher powers of $x^{(i)}$ with a symmetric sum of purely linear factors.

This reduction shows that it suffices to consider multilinear invariants.

Step 2 Extension to $GL_n(\mathbb{C})$: Because $SL_n(\mathbb{C})$ is the subgroup of $GL_n(\mathbb{C})$ defined by $\det(g) = 1$, any $SL_n(\mathbb{C})$ -invariant multilinear polynomial F must transform under the larger group $GL_n(\mathbb{C})$ as a relative invariant:

$$F(g \cdot v^{(1)}, \dots, g \cdot v^{(r)}) = (\det g)^k F(v^{(1)}, \dots, v^{(r)}),$$

where $r = kn$. (If r is not a multiple of n , no non-trivial invariant exists).

To convert this into an absolute invariant, we introduce k sets of n new auxiliary vector variables $y^{(1)}, \dots, y^{(kn)}$, and form k distinct $n \times n$ determinant brackets from them. Because each bracket transforms under $GL_n(\mathbb{C})$ with a factor of $\det(g)$, the product of these brackets B transforms with a factor of $(\det g)^k$. Therefore, the combined polynomial $F \cdot B$ transforms with a factor of $(\det g)^{2k}$.

Step 3 The Cayley Ω -process: Introduce auxiliary vectors $y^{(1)}, \dots, y^{(n)}$ and consider products of brackets in these variables. The Cayley Ω -operator, defined as the determinant of partial derivatives with respect to the coordinates of the $y^{(i)}$, has the property that

when applied to a bracket $[y^{(1)} \dots y^{(n)}]$, it yields a non-zero scalar.

Applying Ω to suitable products $F \cdot B$, where B is a product of brackets in the auxiliary variables, eliminates the auxiliary variables $y^{(i)}$ while preserving $SL_n(\mathbb{C})$ -invariance. This process expresses F as a linear combination of expressions obtained by contracting groups of n vectors using the determinant.

It follows that every multilinear invariant is a linear combination of products of $n \times n$ brackets $[v^{(j_1)} \dots v^{(j_n)}]$. Reversing the polarisation completes the proof. $\square$

2.5 Multilinearity of Symbolic Brackets on Sums of Powers

A key computational advantage of the symbolic method arises when evaluating invariants on forms that admit a decomposition as a sum of powers of linear forms. Such decompositions are closely related to Waring's problem, which seeks minimal expressions of a form as a sum of powers, and to secant varieties of the Veronese embedding, which geometrically encode these decompositions.

Let $F \in \text{Sym}^d(V^*)$ be a homogeneous polynomial admitting a decomposition

$$F = \sum_{v=1}^r \lambda_v L_v^d, \quad \lambda_v \in \mathbb{C}, L_v \in V^*. \quad (1)$$

Because symbolic bracket expressions are multilinear in their arguments, the evaluation of any invariant distributes over such a decomposition. This reduces the computation of high-degree invariants to the evaluation of determinants of linear forms, providing a substantial simplification over expansions in polynomial coefficients.

Proposition 2.6 (Multilinear Evaluation of Symbolic Brackets). *Let Φ be an invariant (or covariant) of degree k , represented symbolically by a bracket monomial $B(\alpha_1, \dots, \alpha_k)$, where each α_i is a symbolic linear form representing F , so that $F = \alpha_1^d = \dots = \alpha_k^d$. Then*

$$\Phi(F) = \sum_{v_1=1}^r \dots \sum_{v_k=1}^r \left(\prod_{i=1}^k \lambda_{v_i} \right) B(L_{v_1}, \dots, L_{v_k}), \quad (2)$$

where $B(L_{v_1}, \dots, L_{v_k})$ denotes the bracket evaluated on the coefficient vectors of the linear forms L_{v_i} .

Proof. Since Φ is homogeneous of degree k , it arises from a symmetric k -linear form $\tilde{\Phi}$ via polarisation:

$$\Phi(F) = \tilde{\Phi}(F, \dots, F).$$

In the symbolic method, $\tilde{\Phi}(F_1, \dots, F_k)$ is represented by assigning independent symbols α_i with $F_i = \alpha_i^d$, and evaluating the bracket monomial $B(\alpha_1, \dots, \alpha_k)$.

Substituting the decomposition $F = \sum_v \lambda_v L_v^d$ into the diagonal evaluation and using multilinearity gives

$$\Phi(F) = \sum_{v_1=1}^r \cdots \sum_{v_k=1}^r \left(\prod_{i=1}^k \lambda_{v_i} \right) \tilde{\Phi}(L_{v_1}^d, \dots, L_{v_k}^d).$$

Because each symbolic bracket is an alternating multilinear function of the coefficient vectors of its linear arguments, evaluating B on $(L_{v_1}^d, \dots, L_{v_k}^d)$ amounts to identifying the symbol α_i with the coefficient vector of L_{v_i} . This yields

$$\tilde{\Phi}(L_{v_1}^d, \dots, L_{v_k}^d) = B(L_{v_1}, \dots, L_{v_k}),$$

which, upon substitution, completes the proof. $\square$

Example: Consider binary forms ($\dim V = 2$) and the invariant $I = [\alpha_1 \alpha_2]^d$. Let

$$F = \lambda_1 L_1^d + \lambda_2 L_2^d, \quad L_1 = x + 2y, \quad L_2 = 3x - y, \quad \lambda_1, \lambda_2 \in \mathbb{C}.$$

Applying Proposition 2.6 gives

$$I(F) = \sum_{i=1}^2 \sum_{j=1}^2 \lambda_i \lambda_j [L_i L_j]^d.$$

The brackets are determinants of the coefficient vectors of the linear forms:

$$\begin{aligned} [L_1 L_1] &= \det \begin{pmatrix} 1 & 2 \\ 1 & 2 \end{pmatrix} = 0, & [L_1 L_2] &= \det \begin{pmatrix} 1 & 2 \\ 3 & -1 \end{pmatrix} = -7, \\ [L_2 L_1] &= \det \begin{pmatrix} 3 & -1 \\ 1 & 2 \end{pmatrix} = 7, & [L_2 L_2] &= \det \begin{pmatrix} 3 & -1 \\ 3 & -1 \end{pmatrix} = 0. \end{aligned}$$

Hence,

$$I(F) = \lambda_1^2 \cdot 0 + \lambda_1 \lambda_2 \cdot (-7)^d + \lambda_2 \lambda_1 \cdot 7^d + \lambda_2^2 \cdot 0 = \lambda_1 \lambda_2 \left((-7)^d + 7^d \right).$$

Remark: For odd d , $(-7)^d + 7^d = 0$, reflecting the antisymmetry of the bracket, while for even d the result is nonzero. This demonstrates how multilinearity and alternating properties of brackets influence the evaluation of invariants.

Geometrically, the invariant is built from the oriented area spanned by the linear factors, illustrating the multilinear reduction from polynomial coefficients to determinants. This approach replaces potentially large expansions in polynomial coefficients with efficient computations in terms of determinants, and is a central mechanism in computational

geometric invariant theory, for example in evaluating invariants on special configurations such as the Hesse pencil [BB08; BBK09].

3 The Old, the New, and the True

The study of invariants, polynomials preserved under coordinate transformations, forms a fundamental link between classical algebra and modern algebraic geometry [Olv99]. Although Hilbert's work shifted the field toward abstract existence results [Hil93], the classical algorithmic techniques of the symbolic method remain highly effective for explicit computation and have seen a modern combinatorial revival [Rot01]. In this chapter, we present the symbolic method as a rigorous algebraic framework. We begin by defining the $SL_2(\mathbb{C})$ -action on binary forms to systematically construct classical objects such as the discriminant, the Hessian covariant, and transvectants [Dol03]. After formulating the general algorithm for invariants and applying it to the binary quartic, we lift the vector space to dimension 3 to study $SL_3(\mathbb{C})$ -actions on ternary forms, providing the algebraic machinery required to study plane cubic curves and the Hesse pencil in subsequent chapters [Muk03].

3.1 Binary Forms and the $SL_2(\mathbb{C})$ -Action

Let $V_{2,d}$ denote the vector space of homogeneous polynomials (binary forms) of degree d in two variables. The central problem of classical invariant theory is to identify polynomial properties of these forms that remain invariant under linear changes of coordinates [Olv99].

We formalise these coordinate transformations by considering the standard left action of the special linear group $SL_2(\mathbb{C})$ on the vector space $\mathbb{C}^2$. For a matrix $g \in SL_2(\mathbb{C})$, the action on the variable vector is given by standard matrix multiplication $x \mapsto gx$. This induces an action on the polynomial ring $\mathbb{C}[x_1, x_2]$. Following Dolgachev [Dol03], we define the induced action of $SL_2(\mathbb{C})$ on the space of binary forms $V_{2,d}$ by:

$$(g \cdot f)(x) = f(g^{-1}x) \tag{3}$$

for any $f \in V_{2,d}$.

A polynomial I in the coefficients of f is called an *invariant* if it is invariant under this action, that is,

$$I(g \cdot f) = I(f)$$

for all $g \in SL_2(\mathbb{C})$ and all $f \in V_{2,d}$.

More generally, we consider polynomials depending on both coefficients and variables of f but also on the original variables x , transforming compatibly with the group action.

A polynomial $C(f, x)$ in both the coefficients of f and the variables x is called a *covariant* if,

$$C(g \cdot f, x) = C(f, g^{-1}x)$$

for all $g \in SL_2(\mathbb{C})$.

While classical invariant theory focused on explicit constructions of these invariants and covariants, the field was profoundly shifted by Hilbert's existential proofs of finite generation [Hil93]. Hilbert's shift from algorithmic construction to abstract algebra laid the groundwork for modern Geometric Invariant Theory (GIT), which generalises such group actions to reductive groups acting on algebraic varieties in order to construct moduli spaces [MFK94; Neu07].

However, for the explicit computation of invariants and covariants, we rely on the symbolic method developed in the previous sections. The connection between the abstract $SL_2(\mathbb{C})$ -action and the symbolic method lies in the transformation of the determinant. Recall that the bracket of two symbolic linear forms $a_x = a_1x_1 + a_2x_2$ and $b_x = b_1x_1 + b_2x_2$ is defined as the determinant of their coefficients: $[ab] = a_1b_2 - a_2b_1$. Under $g \in GL_2(\mathbb{C})$, this bracket transforms as:

$$[ab] \mapsto \det(g)[ab]. \quad (4)$$

Because $\det(g) = 1$ by the definition of the special linear group, the bracket $[ab]$ is therefore invariant under $SL_2(\mathbb{C})$. Consequently, any polynomial expression constructed entirely from contracted symbolic brackets evaluates to an $SL_2(\mathbb{C})$ -invariant, while expressions containing uncontracted factors (such as a_x) produce covariants.

3.2 The Discriminant in Low Degree

To illustrate the symbolic method, we construct the simplest non-trivial invariant: the discriminant of a binary quadratic. Using standard coordinates, fundamental is the discriminant, which detects multiple roots. For a quadratic

$$f(x, y) = ax^2 + 2bxy + cy^2 \quad (5)$$

we define

$$\Delta(f) \doteq ac - b^2. \quad (6)$$

Proposition 3.1. *The discriminant Δ is an $SL_2(\mathbb{C})$ -invariant of degree 2.*

Proof. Write $g = \begin{pmatrix} p & q \\ r & s \end{pmatrix} \in SL_2(\mathbb{C})$. Then by the group action defined in Section ??,

$$\begin{aligned} (g \cdot f)(x, y) &= f((x, y) \cdot g) \\ &= a(px + ry)^2 + 2b(px + ry)(qx + sy) + c(qx + sy)^2. \end{aligned}$$

Collecting coefficients gives new coefficients a', b', c' depending polynomially on a, b, c and p, q, r, s . A direct computation shows that

$$a'c' - (b')^2 = (ps - qr)^2(ac - b^2) = ac - b^2,$$

since $\det(g) = 1$. Thus $\Delta(g \cdot f) = \Delta(f)$ (see [Dol03, §3.1]). □

Whilst this direct computation is manageable in low degree, it becomes increasingly complicated for higher degrees. This is precisely where we appeal to the symbolic method. By symbolically representing the quadratic as $f \sim \alpha_x^2 \sim \beta_x^2$, the coefficients evaluate to $\alpha_1^2 = a$, $\alpha_1\alpha_2 = b$, and $\alpha_2^2 = c$. One fully contracted invariant bracket polynomial is $\frac{1}{2}[\alpha\beta]^2$, which instantly expands to:

$$\frac{1}{2}(\alpha_1\beta_2 - \alpha_2\beta_1)^2 = \alpha_1^2\beta_2^2 - 2(\alpha_1\alpha_2)(\beta_1\beta_2) + \alpha_2^2\beta_1^2 = ac - b^2 = \Delta(f). \quad (7)$$

The symbolic combinatorics bypass the matrix transformations entirely [GY03].

For higher degrees, manually computing transformations becomes a massive algebraic undertaking. We record the cubic case here to illustrate the combinatorial explosion that the symbolic method was designed to solve.

A general cubic is

$$f(x, y) = ax^3 + 3bx^2y + 3cxy^2 + dy^3.$$

The discriminant is the degree-4 polynomial

$$\Delta(f) = a^2d^2 - 6abcd + 4ac^3 + 4b^3d - 3b^2c^2.$$

It vanishes exactly when f has a multiple root in $\mathbb{P}^1$. One checks by a direct change of variables that Δ is invariant under $SL_2(\mathbb{C})$. Symbolically, this expression is captured by a bracket monomial $c \cdot [\alpha\beta]^2[\gamma\delta]^2[\alpha\gamma][\beta\delta]$ (where c is a normalising scalar).

3.3 The Hessian Covariant for Binary Cubics

Having seen that fully contracted bracket polynomials correspond to invariants, we now show that allowing uncontracted symbolic variables naturally produces covariants. A classical and geometrically significant example is the Hessian.

Let $f \in V_{2,d}$ be a binary form. The *Hessian* of f is the binary form

$$\text{Hess}(f)(x, y) \doteq \det \begin{pmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} \end{pmatrix}. \quad (8)$$

Since f has degree d in the variables and is linear in its coefficients, the second derivatives have degree $d - 2$ in (x, y) . The determinant multiplies two such derivatives together, meaning $\text{Hess}(f)$ has degree $2d - 4$ in (x, y) and is homogeneous of degree 2 in the coefficients of f .

Proposition 3.2. *The Hessian is a covariant of degree 2 and order $2d - 4$:*

$$\text{Hess}(g \cdot f; (x, y) \cdot g) = \text{Hess}(f; x, y) \quad \text{for all } g \in SL_2(\mathbb{C}).$$

Proof. Although this can be proved using the chain rule to the second derivatives (which shows the Hessian transforms by a factor of $(\det g)^2 = 1$, as detailed in [Dol03, §2.3]), the symbolic method provides a more conceptual derivation.

Assign two equivalent independent symbolic representations to the form: $f \sim \alpha_x^d \sim \beta_x^d$, where $\alpha_x = \alpha_1 x + \alpha_2 y$. Computing the formal second derivatives,

$$\frac{\partial^2 f}{\partial x^2} = d(d-1)\alpha_1^2\alpha_x^{d-2}, \quad \frac{\partial^2 f}{\partial y^2} = d(d-1)\alpha_2^2\alpha_x^{d-2}, \quad \frac{\partial^2 f}{\partial x \partial y} = d(d-1)\alpha_1\alpha_2\alpha_x^{d-2}.$$

To avoid trivial cancellation in the determinant, we introduce an independent symbol β for the second column of the matrix. Substituting these into the definition of $\text{Hess}(f)$, we obtain:

$$\begin{aligned} \text{Hess}(f) &= \left(d(d-1)\alpha_1^2\alpha_x^{d-2} \right) \left(d(d-1)\beta_2^2\beta_x^{d-2} \right) - \left(d(d-1)\alpha_1\alpha_2\alpha_x^{d-2} \right) \left(d(d-1)\beta_1\beta_2\beta_x^{d-2} \right) \\ &= d^2(d-1)^2\alpha_x^{d-2}\beta_x^{d-2}(\alpha_1^2\beta_2^2 - \alpha_1\alpha_2\beta_1\beta_2). \end{aligned}$$

Because α and β are auxiliary symbols representing the exact same form f , this expression must be symmetric under the interchange of α and β . Swapping the symbols and symmetrising yields:

$$\begin{aligned} \text{Hess}(f) &= \frac{1}{2}d^2(d-1)^2\alpha_x^{d-2}\beta_x^{d-2}(\alpha_1^2\beta_2^2 - 2\alpha_1\alpha_2\beta_1\beta_2 + \beta_1^2\alpha_2^2) \\ &= \frac{1}{2}d^2(d-1)^2\alpha_x^{d-2}\beta_x^{d-2}(\alpha_1\beta_2 - \alpha_2\beta_1)^2. \end{aligned}$$

Recognising the squared term as the bracket $[\alpha\beta]^2$, we conclude that:

$$\text{Hess}(f) = \frac{1}{2}d^2(d-1)^2[\alpha\beta]^2\alpha_x^{d-2}\beta_x^{d-2}. \quad (9)$$

Since this expression consists of an invariant bracket factor multiplied by linear forms, it defines a covariant. $\square$

Let

$$f(x, y) = ax^4 + 4bx^3y + 6cx^2y^2 + 4dxy^3 + ey^4$$

be a binary quartic ($d = 4$). The symbolic formula yields $\text{Hess}(f) \propto [\alpha\beta]^2 \alpha_x^2 \beta_x^2$, which shows it is a covariant of order $2(4) - 4 = 4$. A direct coordinate computation yields:

$$\text{Hess}(f)(x, y) = Ax^4 + 4Bx^3y + 6Cx^2y^2 + 4Dxy^3 + Ey^4,$$

where

$$A = 12(ac - b^2),$$

$$B = 6(ad - bc),$$

$$C = 12ae + 6bd - 18c^2,$$

$$D = 6(be - cd),$$

$$E = 12(ce - d^2).$$

Thus the Hessian associates to f another quartic whose coefficients are quadratic polynomials in a, b, c, d, e . The span of f and $\text{Hess}(f)$ plays an important role in the invariant theory of quartics.

In the higher-dimensional setting of ternary forms, the Hessian completely governs the geometry of plane cubic curves and their inflection points, forming the foundation of the Hesse pencil [AD09]. The algebraic properties of this pencil remain an active area of research in modern algebraic geometry, particularly in classifying families of genus one curves and their moduli spaces [Roc25].

3.4 Transvectants

Transvectants provide a systematic differential method for constructing covariants and invariants. We begin with an analytical definition.

Let $f \in V_{2,m}$ and $g \in V_{2,n}$ be binary forms of degrees m and n in the variables (x, y) . For an integer $0 \leq r \leq \min\{m, n\}$, the r -th transvectant of f and g is defined as:

$$(f, g)_r(x, y) \doteq \sum_{i=0}^r (-1)^i \binom{r}{i} \frac{\partial^r f}{\partial x^{r-i} \partial y^i} \frac{\partial^r g}{\partial x^i \partial y^{r-i}}. \quad (10)$$

The transvectant $(f, g)_r$ is a binary form of degree $m + n - 2r$ in (x, y) , and it is bilinear in

the coefficients of f and g .

This construction can be interpreted using the Cayley Ω -operator introduced in previous section. By introducing auxiliary variables (u, v) for g , we can rewrite the transvectant using the invariant differential operator $\Omega = \frac{\partial^2}{\partial x \partial v} - \frac{\partial^2}{\partial y \partial u}$. Up to a normalising scalar, the transvectant is exactly the application of this operator r times, evaluated at the diagonal:

$$(f, g)_r(x, y) \propto \Omega^r(f(x, y)g(u, v)) \Big|_{u=x, v=y}. \quad (11)$$

Proposition 3.3. *If f and g are covariants of degrees d_f and d_g , then their r -th transvectant $(f, g)_r$ is a covariant of degree $d_f + d_g$ and order $m + n - 2r$.*

Proof. We prove this rigorously using the symbolic method. Assign symbolic representations $f \sim \alpha_x^m$, $g \sim \beta_u^n$, where $\alpha_x = \alpha_1 x + \alpha_2 y$ and $\beta_u = \beta_1 u + \beta_2 v$.

Applying the Cayley operator once to the product $\alpha_x^m \beta_u^n$ gives

$$\begin{aligned} \Omega(\alpha_x^m \beta_u^n) &= \left(\frac{\partial^2}{\partial x \partial v} - \frac{\partial^2}{\partial y \partial u} \right) (\alpha_1 x + \alpha_2 y)^m (\beta_1 u + \beta_2 v)^n \\ &= mn(\alpha_1 \beta_2 - \alpha_2 \beta_1) \alpha_x^{m-1} \beta_u^{n-1} \\ &= mn[\alpha\beta] \alpha_x^{m-1} \beta_u^{n-1}. \end{aligned}$$

Iterating this operator r times sequentially pulls down exactly r factors of the bracket:

$$\Omega^r(\alpha_x^m \beta_u^n) = \frac{m!}{(m-r)!} \frac{n!}{(n-r)!} [\alpha\beta]^r \alpha_x^{m-r} \beta_u^{n-r}.$$

Finally, evaluating at the diagonal by setting $u = x$ and $v = y$ provides the symbolic expression for the transvectant:

$$(f, g)_r \propto [\alpha\beta]^r \alpha_x^{m-r} \beta_x^{n-r}. \quad (12)$$

By the fundamental theorem of the symbolic method (Section ??), this expression is constructed entirely from an invariant contracted bracket $[\alpha\beta]^r$ multiplied by uncontracted covariant linear forms α_x and β_x . Therefore, $(f, g)_r$ transforms as a covariant of order $(m-r) + (n-r) = m + n - 2r$. $\square$

Let $f \in V_{2,d}$. The second transvectant $(f, f)_2$ evaluates to:

$$(f, f)_2 = \binom{2}{0} \frac{\partial^2 f}{\partial x^2} \frac{\partial^2 f}{\partial y^2} - \binom{2}{1} \frac{\partial^2 f}{\partial x \partial y} \frac{\partial^2 f}{\partial x \partial y} + \binom{2}{2} \frac{\partial^2 f}{\partial y^2} \frac{\partial^2 f}{\partial x^2}.$$

Combining these terms yields exactly twice the determinant of the Hessian matrix. Thus:

$$(f, f)_2 = 2 \text{Hess}(f).$$

This identifies the Hessian as the second self-transvectant of f , completely explaining its naturality from both the differential and representation-theoretic perspectives.

3.5 The General Algorithm for Invariants

The preceding constructions of the discriminant, the Hessian, and general transvectants all rely on a singular underlying principle: $SL_2(\mathbb{C})$ -invariants are in bijection with fully contracted bracket monomials. The First Fundamental Theorem of Invariant Theory rigorously formalizes this, stating that *every* invariant of a binary form can be generated this way.

This reduces the daunting algebraic problem of finding invariants into a purely combinatorial algorithm, famously championed by Grace and Young [GY03] and later given rigorous combinatorial foundations by Rota [RS86; DRS74].

The following is an algorithm for constructing invariants I of degree k for a binary form $f \in V_{2,d}$ of degree d :

1. Assign k independent, equivalent symbolic representations to the form, $f \sim \alpha_x^d \sim \beta_x^d \sim \gamma_x^d \sim \dots$, generating k distinct auxiliary symbols.
2. Form a product of symbolic brackets (e.g., $[\alpha\beta]$, $[\alpha\gamma]$, etc.) such that the expression is fully contracted (no x or y variables remain).
3. As the invariant has degree k in the coefficients of f , when constructing the brackets, the k symbols must appear exactly d times across all brackets.
4. Expand the brackets algebraically and substitute the symbolic monomials back into the actual coefficients of f .

Proposition 3.4. *A non-trivial invariant I of degree k for a binary form $f \in V_{2,d}$ can only exist if kd is even.*

Proof. By the algorithm, the invariant is a product of brackets. Each bracket $[\alpha\beta]$ absorbs exactly two symbol appearances (one α and one β). Because there are k symbols and each must appear exactly d times, the total number of symbol appearances is kd . Since each bracket has two occurrences, the total number of brackets must be $w = \frac{kd}{2}$. For w to be an integer, kd must be even. $\square$

Remark 3.5 (The Graph-Theoretic Perspective). This algorithm shows us that the symbolic method translates algebraic geometry into graph theory. A bracket monomial can be viewed as a multigraph where the k auxiliary symbols are vertices, and each bracket $[\alpha\beta]$

is an undirected edge connecting vertex α and vertex β . The regularity condition implies that an invariant corresponds to a d -regular graph on k vertices [Rot01; Gro03].

Historically, this algorithmic generation of invariants was known as Gordan's method, which was the prequel to Gordan's Theorem in 1868. He proved that the ring of invariants for any binary form is finitely generated. However, the combinatorial nightmare of finding the minimal generating set was a point of crisis in the field. When Hilbert proved his finiteness theorem using abstract algebra rather than symbolic brackets, classical algorithmic invariant theory famously declined [Hil93; Cri88].

It was not until the late 20th century when Rota revitalized the theory by connecting bracket algebra to modern combinatorics [Rot01]. Meanwhile, the advent of computer algebra programs made the bracket expansions more trivial. Today, algorithmic invariant theory is a highly active field, bridging classical geometry with modern software implementations in systems like OSCAR [Stu08; DK15; DRS25].

3.6 A Worked Example: The Invariant I of the Quartic

We now demonstrate the algorithm in full by constructing the classical invariants of the binary quartic. Let $f \in V_{2,4}$ be the general quartic with binomial coefficients:

$$f(x, y) = ax^4 + 4bx^3y + 6cx^2y^2 + 4dxy^3 + ey^4. \quad (13)$$

Its invariants naturally define the algebras associated to genus one curves (elliptic curves), serving as the lowest-degree bridge between invariant theory and algebraic geometry [Fis19].

By Gordan's Theorem, the invariant ring of the binary quartic is generated by exactly two algebraically independent invariants: a degree-2 invariant I , and a degree-3 invariant J . Conceptually, these invariants correspond to the unique one-dimensional SL_2 -equivariant contractions of the tensor products $\text{Sym}^4(\mathbb{C}^2)^\vee \otimes \text{Sym}^4(\mathbb{C}^2)^\vee \rightarrow \mathbb{C}$ and $\text{Sym}^4(\mathbb{C}^2)^\vee \otimes \text{Sym}^4(\mathbb{C}^2)^\vee \otimes \text{Sym}^4(\mathbb{C}^2)^\vee \rightarrow \mathbb{C}$, respectively [Dol03, §3.5].

We compute I using the symbolic algorithm.

1. We want an invariant of degree $k = 2$, so introduce two equivalent auxiliary symbols, α and β , such that $f \sim \alpha_x^4 \sim \beta_x^4$. Because the quartic has degree $d = 4$, each symbol must appear exactly 4 times in the final bracket monomial.
2. Since invariants contain no free (x, y) variables, we must absorb all 8 symbol instances into fully contracted brackets. In graph-theory terms, we seek a 4-regular graph on 2 vertices. The only possible connected graph is four parallel edges connecting vertex α to vertex β , yielding the unique bracket monomial:

$$[\alpha\beta]^4.$$

3. We expand the bracket algebraically using the binomial theorem:

$$\begin{aligned} [\alpha\beta]^4 &= (\alpha_1\beta_2 - \alpha_2\beta_1)^4 \\ &= \alpha_1^4\beta_2^4 - 4\alpha_1^3\alpha_2\beta_1\beta_2^3 + 6\alpha_1^2\alpha_2^2\beta_1^2\beta_2^2 - 4\alpha_1\alpha_2^3\beta_1^3\beta_2 + \alpha_2^4\beta_1^4. \end{aligned}$$

4. Now replace the symbolic monomials with the actual coefficients of the quartic. The symbolic equivalences show,

$$\alpha_1^4 = \beta_1^4 = a, \quad \alpha_1^3\alpha_2 = \beta_1^3\beta_2 = b, \quad \alpha_1^2\alpha_2^2 = \beta_1^2\beta_2^2 = c, \quad \alpha_1\alpha_2^3 = \beta_1\beta_2^3 = d, \quad \alpha_2^4 = \beta_2^4 = e.$$

Substituting these into our expanded bracket,

$$\begin{aligned} [\alpha\beta]^4 &\mapsto a(e) - 4b(d) + 6c(c) - 4d(b) + e(a) \\ &= 2ae - 8bd + 6c^2. \end{aligned}$$

Finally, dividing by the normalisation constant 2, we recover the classical degree-2 invariant:

$$I = ae - 4bd + 3c^2. \quad (14)$$

The procedure is identical for the degree-3 invariant J . This time we require a 4-regular graph on 3 vertices (α, β, γ) . The unique solution is a triangle with doubled edges between every pair of vertices. This immediately yields the symbolic representation:

$$J \propto [\alpha\beta]^2[\beta\gamma]^2[\gamma\alpha]^2. \quad (15)$$

Expanding this degree-6 bracket polynomial and substituting the coefficients yields the classical result $J = ace + 2bcd - ad^2 - eb^2 - c^3$. The uniqueness of the graph structures for I and J perfectly reflects the fact that the graded pieces $I_{4,2}$ and $I_{4,3}$ of the invariant ring are one-dimensional.

3.7 Lifting to Dimension 3

Let $V_{3,d}$ denote the vector space of ternary forms of degree d . We define the underlying space as $V = \mathbb{C}^3$ with coordinates $x = (x_1, x_2, x_3)$. A ternary form $f(x) \in V_{3,d}$ is symbolised by introducing vectors $\alpha = (\alpha_1, \alpha_2, \alpha_3) \in V^*$ such that,

$$f(x) \sim (\alpha_1x_1 + \alpha_2x_2 + \alpha_3x_3)^d = (\alpha \cdot x)^d. \quad (16)$$

Similar to the binary case, we must introduce multiple equivalent copies of the symbolic representation to build non-trivial invariants. We write $f(x) \sim (\alpha \cdot x)^d \sim (\beta \cdot x)^d \sim (\gamma \cdot x)^d$, and so on.

According to the First Fundamental Theorem of Invariant Theory for $SL_3(\mathbb{C})$, all invariants

of a ternary form can be expressed as polynomials in the fundamental 3×3 determinant bracket [GY03]:

$$[\alpha\beta\gamma] = \det \begin{pmatrix} \alpha_1 & \alpha_2 & \alpha_3 \\ \beta_1 & \beta_2 & \beta_3 \\ \gamma_1 & \gamma_2 & \gamma_3 \end{pmatrix}. \quad (17)$$

This generalisation from a 2×2 bracket to a 3×3 bracket was systematically developed by Clebsch in 1861, who recognized that the geometry of plane curves could be entirely encoded in these symbols.

We turn to constructing invariants of degree k for a quantic form of degree d with the following algorithm. When $n = 3$, this generates the invariants of ternary forms [Stu08]:

1. Each of the k symbolic vectors $\alpha^{(1)}, \dots, \alpha^{(k)}$ represents a copy of f , so each must appear exactly d times in the final expression to perfectly absorb the x -dependence.
2. Form products of $n \times n$ brackets using these k symbols. Since each bracket requires n symbols, the total number of brackets required is $w = (k \cdot d)/n$. (If kd is not divisible by n , no invariant of degree k exists).
3. Sum over all permutations of the k symbols to create a fully symmetric bracket polynomial. This ensures the expression is independent of the arbitrary assignment of symbols to copies of f .
4. Because 3×3 brackets are not algebraically independent, apply the fundamental Plücker syzygy to reduce the expression to a basis of standard tableaux:

$$[\alpha\beta\gamma][\delta\varepsilon\zeta] = [\alpha\beta\delta][\gamma\varepsilon\zeta] - [\alpha\beta\varepsilon][\gamma\delta\zeta] + [\alpha\beta\zeta][\gamma\delta\varepsilon]. \quad (18)$$

5. Finally, replace the symbolic products with the actual coefficients of f via the Veronese identification.

3.8 Plane Cubics, the Hesse Pencil, and the Geometry of S and T

A ternary cubic form $f(x, y, z) \in V_{3,3}$ defines a cubic curve in the projective plane $\mathbb{P}^2$. The classical problem of classifying these curves up to projective equivalence is equivalent to constructing the moduli space of elliptic curves. We overcome dealing with the 10 coefficients of a general cubic by invoking a fundamental geometric theorem: any smooth plane cubic can be transformed via an $SL_3(\mathbb{C})$ coordinate change into the *Hesse canonical form*:

$$f_\lambda(x, y, z) = x^3 + y^3 + z^3 + 6\lambda xyz, \quad (19)$$

where $\lambda \in \mathbb{C}$ is a parameter. This 1-parameter family of curves is known as the Hesse pencil [AD09].

Following Mukai [Muk03, p. 430], the invariant ring of ternary cubics is generated by

the Aronhold invariant S (degree 4) and the Brioschi invariant T (degree 6). When we evaluate it on the Hesse form, these invariants reduce to univariate polynomials in λ ,

$$S(f_\lambda) = -\lambda + \lambda^4, \quad (20)$$

$$T(f_\lambda) = 1 - 20\lambda^3 - 8\lambda^6. \quad (21)$$

The discriminant of the cubic, which vanishes precisely when the curve becomes singular (e.g., forming a nodal or cuspidal cubic), is given by the algebraic syzygy,

$$\Delta(f_\lambda) = T(f_\lambda)^2 - 64S(f_\lambda)^3 = (1 + 8\lambda^3)^3. \quad (22)$$

This parametrisation reveals the geometric significance of the invariants S and T , which act as detectors for the curve's symmetries:

- **When $S = 0$:** The cubic is called *equianharmonic*. From the formula, $S = 0$ implies $\lambda \in \{0, 1, \zeta, \zeta^2\}$ (where ζ is a primitive cube root of unity). The case $\lambda = 0$ recovers the Fermat cubic $x^3 + y^3 + z^3 = 0$. Such curves have an exceptionally large automorphism group of order 54, and their j -invariant is 0.
- **When $T = 0$:** The cubic is called *harmonic*. Its j -invariant is exactly 1728, and the curve admits an automorphism group of order 24.

Whence, S and T are the fundamental geometric detectors of plane curves. A full geometric exploration of these properties, and their role in the construction of moduli spaces, will be the focus of the subsequent chapters.

4 The Bigger Picture

In this section, we focus on the geometric application of the symbolic method. We can characterise the loci defined by the vanishing of classical invariants, as well as use the brackets of the symbolic method encode the projective geometry of underlying forms.

4.1 Geometric Loci of Quartic Invariants

The geometric meaning of the vanishing invariants I and J is seen after carefully evaluating them on the moduli space of four points on $\mathbb{P}^1$.

By the fundamental theorem of projective geometry, any four distinct points on $\mathbb{P}^1$ can be mapped by an element of $\text{PSL}_2(\mathbb{C})$ to the points $0, 1, \infty$, and $\lambda \in \mathbb{C} \setminus \{0, 1\}$ is the cross-ratio [Dol03], [Muk03]. The corresponding binary quartic with these roots may be written, after dehomogenisation as,

$$f(x, y) = xy(x - y)(x - \lambda y) = x^3y - (\lambda + 1)x^2y^2 + \lambda xy^3.$$

Comparing this to the general binary quartic

$$ax^4 + 4bx^3y + 6cx^2y^2 + 4dxy^3 + ey^4$$

, we identify the coefficient as

$$a = 0, \quad b = \frac{1}{4}, \quad c = -\frac{\lambda + 1}{6}, \quad d = \frac{\lambda}{4}, \quad e = 0,$$

after appropriate normalisation. We now evaluate the classical invariants I and J as functions of the cross-ratio λ :

$$\begin{aligned} I(\lambda) &= ae - 4bd + 3c^2 \\ &= 0 - 4\left(\frac{1}{4}\right)\left(\frac{\lambda}{4}\right) + 3\left(-\frac{\lambda + 1}{6}\right)^2 \\ &= -\frac{\lambda}{4} + \frac{\lambda^2 + 2\lambda + 1}{12} \\ &= \frac{\lambda^2 - \lambda + 1}{12}. \end{aligned}$$

This expression is symmetric under the classical action of the cross-ratio group, reflecting the intrinsic symmetry of the configuration of four points on $\mathbb{P}^1$. In particular, the vanishing of $I(\lambda)$ corresponds to special configurations of points where the cross-ratio satisfies

$$\lambda^2 - \lambda + 1 = 0,$$

which are precisely the equiharmonic configurations. Thus, the invariant I defines a geometric locus in the moduli space of four points on $\mathbb{P}^1$.

Similarly, we compute

$$\begin{aligned} J(\lambda) &= \det \begin{pmatrix} a & b & c \\ b & c & d \\ c & d & e \end{pmatrix} = \det \begin{pmatrix} 0 & 1/4 & c \\ 1/4 & c & \lambda/4 \\ c & \lambda/4 & 0 \end{pmatrix} \\ &= 2bcd - c^3 = 2\left(\frac{1}{4}\right)\left(-\frac{\lambda + 1}{6}\right)\left(\frac{\lambda}{4}\right) - \left(-\frac{\lambda + 1}{6}\right)^3 \\ &= \frac{(\lambda + 1)(\lambda - 2)(2\lambda - 1)}{432}. \end{aligned}$$

As a consequence, we identify the vanishing loci:

- **The locus $I = 0$:** The roots of $I(\lambda) = 0$ are $\lambda = e^{\pm i\pi/3} = \frac{1 \pm i\sqrt{3}}{2}$. These are the *equianharmonic* cross-ratios. Geometrically, the four points form a regular tetrahedron on the Riemann sphere. The associated elliptic curve $y^2 = f(x, 1)$ admits an extra automorphism of order 6 [Fis19].

- **The locus $J = 0$:** The roots of $J(\lambda) = 0$ are $\lambda \in \{-1, \frac{1}{2}, 2\}$. These are the *harmonic* cross-ratios. Geometrically, the four points are harmonic conjugates. The corresponding elliptic curve admits an extra automorphism of order 4 [Fis19].

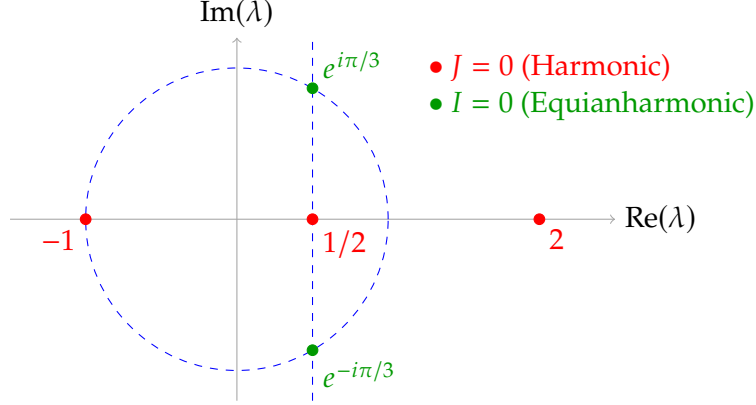


Figure 1: The configuration of the cross-ratio λ in the complex plane. The equianharmonic cross-ratios ($I = 0$) lie precisely at the intersections of the unit circle $|\lambda| = 1$ and the vertical line $\text{Re}(\lambda) = \frac{1}{2}$.

4.2 Geometric Loci of Ternary Cubics via the Hesse Pencil

To understand the geometric meaning of the Aronhold invariant S (degree 4) and the Brioschi invariant T (degree 6) for ternary cubics, we evaluate them on the *Hesse normal form*. The Hesse pencil provides a parametrisation of a family of plane cubic curves, which may be interpreted as a slice through the moduli space of elliptic curves. Any smooth plane cubic can be represented in $\mathbb{P}^2$ as a member of the Hesse pencil [Muk03],

$$f_\lambda(x, y, z) = x^3 + y^3 + z^3 + 6\lambda xyz. \quad (23)$$

By classical invariant theory, substituting the coefficients of F_λ into the symbolic expressions for S and T yields elegant univariate polynomials:

$$S(f_\lambda) = -\lambda + \lambda^4, \quad (24)$$

$$T(f_\lambda) = 1 - 20\lambda^3 - 8\lambda^6. \quad (25)$$

From these explicit parametrisations, we can classify the geometry of the vanishing loci:

- **The locus $S = 0$:** The equation $-\lambda + \lambda^4 = 0$ has solutions $\lambda = 0$ and $\lambda^3 = 1$. The curve is called *equianharmonic*. When $\lambda = 0$, the curve is exactly the Fermat cubic $x^3 + y^3 + z^3 = 0$. The roots $\lambda^3 = 1$ yield curves that are projectively equivalent to the Fermat cubic. Such curves have an exceptionally large automorphism group of order 54, and their absolute j -invariant is 0.
- **The locus $T = 0$:** The cubic is called *harmonic*. Its j -invariant is exactly 1728, and the

curve admits an extra automorphism of order 24.

- **The Discriminant Locus $\Delta = 0$:** The discriminant of a ternary cubic is given by the syzygy $\Delta = T^2 - 64S^3$. Computing this for the Hesse pencil gives:

$$\Delta(\lambda) = (1 - 20\lambda^3 - 8\lambda^6)^2 - 64(-\lambda + \lambda^4)^3 = (1 + 8\lambda^3)^3.$$

Thus, $\Delta = 0$ if and only if $\lambda^3 = -1/8$ (e.g., $\lambda = -1/2$). Substituting $\lambda = -1/2$ into the pencil gives $x^3 + y^3 + z^3 - 3xyz = 0$, which famously factors as:

$$(x + y + z)(x^2 + y^2 + z^2 - xy - yz - xz) = 0. \quad (26)$$

This factorisation indicates a reducible curve formed by the union of a line and a conic. Thus, the condition $\Delta = 0$ correctly and algebraically detects singular, degenerate fibers in the moduli space.

The fact that $S = 0$ characterises the Fermat cubic and its projective equivalents is not just an algebraic coincidence. To see why the S -invariant detects this geometry so precisely, we must examine its structural formulation via the symbolic method.

The Aronhold invariant S requires $k = 4$ symbols $(\alpha, \beta, \gamma, \delta)$ and $w = 4$ brackets to satisfy the degree constraints. The unique fully symmetric non-vanishing bracket polynomial is:

$$S(f) \propto [\alpha\beta\gamma][\beta\gamma\delta][\gamma\delta\alpha][\delta\alpha\beta]. \quad (27)$$

Returning to Remark 3.5, we reinterpret our calculations using graph theory. We can represent the structure of $S(f)$ as a topological graph where each of the 4 symbols is a vertex, and each of the 4 brackets is a triangular face connecting three vertices.

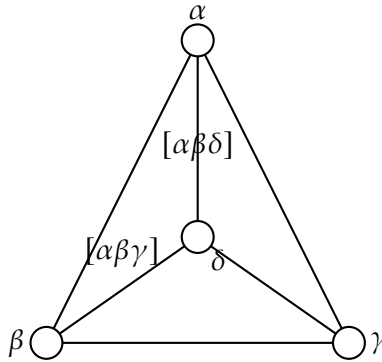


Figure 2: The incidence structure of the Aronhold S -invariant forms a topological tetrahedron (the complete graph K_4). Every symbol (vertex) is incident to exactly three brackets (faces), satisfying the requirement that the ternary cubic has degree $d = 3$.

The symbolic representation of $S(f)$ perfectly matches the structure of a tetrahedron. Every vertex belongs to exactly three faces, matching the condition that each symbol must

be used exactly $d = 3$ times. We can leverage this multilinear combinatorial structure to rigorously prove that S vanishes on the Fermat cubic.

Proposition 4.1. *The Aronhold invariant S vanishes on the Fermat cubic $f(x_1, x_2, x_3) = x_1^3 + x_2^3 + x_3^3$.*

Proof. A generic form can be written as a sum of powers of linear forms, representation known as a Waring decomposition [CCG12]. Notice that the Fermat cubic is already presented as a sum of three cubes: $f(x) = \ell_1^3 + \ell_2^3 + \ell_3^3$, where $\ell_1 = (1, 0, 0)$, $\ell_2 = (0, 1, 0)$, and $\ell_3 = (0, 0, 1)$.

Because the bracket polynomial $S(f)$ is multilinear in its four symbolic inputs, Lemma 5.9 (following Böhning and Graf von Bothmer [BBK09]) allows us to distribute the evaluation across the linear forms. We replace each of the four symbolic slots $(\alpha, \beta, \gamma, \delta)$ with the sum $\sum_{i=1}^3 \ell_i$, yielding a sum over $3^4 = 81$ terms,

$$S(f) = \sum_{i,j,k,m \in \{1,2,3\}} [\ell_i \ell_j \ell_k][\ell_j \ell_k \ell_m][\ell_k \ell_m \ell_i][\ell_m \ell_i \ell_j]. \quad (28)$$

The determinant bracket $[\ell_a \ell_b \ell_c]$ vanishes identically if any two vectors are the same. Look at the first bracket: $[\ell_i \ell_j \ell_k]$. For this to be non-zero, i, j , and k must be distinct. Since our index pool is only $\{1, 2, 3\}$, they must be a permutation of $1, 2, 3$.

This leaves the fourth index, m . But m must take a value from $\{1, 2, 3\}$. Therefore, m must equal either i, j , or k . By the Pigeonhole Principle, no matter which value m takes, one of the three remaining brackets in the product will contain a repeated vector and vanish. For instance, if $m = i$, the fourth bracket becomes $[\ell_i \ell_i \ell_j] = 0$.

Thus, every single one of the 81 terms vanishes, elegantly proving that $S(f) = 0$. $\square$

The geometric implication is that the Fermat cubic is apolar to its own Hessian, placing it in the unique orbit of cubics that can be written as a sum of three cubes with no mixed term [Dol03].

4.3 Generic Rank and Secant Varieties

Just as the Aronhold invariant of a ternary cubic is intrinsically linked to its Waring decomposition, the invariants of quantic forms encode deep geometric data about their symmetric tensor rank [Com+08].

The *Waring rank* (or symmetric tensor rank) of a quantic form f of degree d is the minimum integer r such that f can be expressed as a sum of d -th powers of r linear forms:

$$f = \sum_{j=1}^r \ell_j^d. \quad (29)$$

By Alexander and Hirschowitz's theorem, the generic rank of a quantic form of degree d is given by $r = \lceil \frac{1}{n} \binom{n+d-1}{d} \rceil$, with a small number of known exceptions [CCG12].

Evaluating this dimension count for the two primary spaces studied in this chapter yields:

- For a binary quartic ($n = 2, d = 4$), the generic rank is $\lceil \frac{1}{2} \binom{5}{4} \rceil = \lceil 2.5 \rceil = 3$.
- For a ternary cubic ($n = 3, d = 3$), the generic rank is $\lceil \frac{1}{3} \binom{5}{3} \rceil = \lceil 3.33 \rceil = 4$.

Geometrically, the forms of rank 1 (pure powers $f = \ell^d$) parametrise the *Veronese variety* $X = v_d(\mathbb{P}^{n-1})$ embedded in $\mathbb{P}^{\binom{n+d-1}{d}-1}$. A form of rank r is a linear combination of r points on X , meaning it lies in the linear span of those points. The Zariski closure of the union of all such r -dimensional secant planes forms the r -th *secant variety* $\sigma_r(X)$.

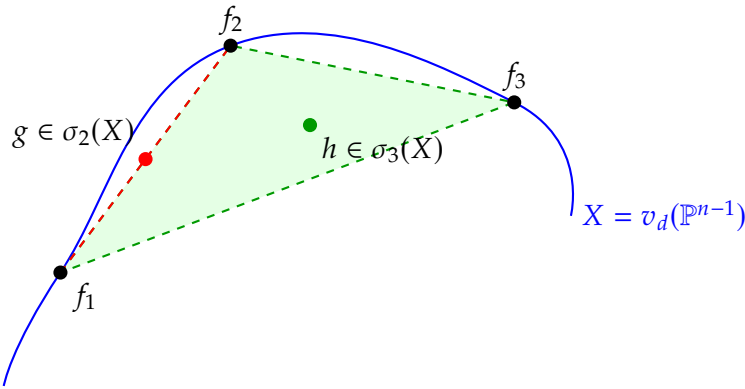


Figure 3: Geometric interpretation of Waring rank. Forms of rank 1 (f_i) lie precisely on the Veronese variety X . A form g of rank 2 lies on a secant line (red). A form h of rank 3 lies on a secant plane (green). Symbolic invariants act as defining algebraic equations that detect when a general form degenerates onto these lower-dimensional secant varieties.

When we compute bracket invariants via the symbolic method, evaluating these invariants on a form expressed in its Waring decomposition forces a significant reduction in messy computations. Consider an invariant I constructed from k auxiliary symbols. When evaluated on a form of Waring rank r , the multilinearity of the symbolic method allows us to distribute the k symbolic slots across the r linear forms $\ell_1, \dots, \ell_r$.

If $k > r$, the Pigeonhole Principle dictates that at least two symbols must be assigned the exact same linear form ℓ_i . Because the invariant is constructed from $n \times n$ determinant brackets, any bracket containing both of these coincident symbols identically vanishes.

This combinatorial constraint predicts the geometric vanishing loci for both binary quartics and ternary cubics,

- **The Binary Quartic:** As computed above, the generic rank is $r = 3$. The invariant J has degree $k = 3$ symbols and forms a K_3 triangle graph. If a quartic degenerates to rank $r = 2$, we distribute $k = 3$ symbols among 2 linear forms. A collision is forced.

Since every pair of symbols in J shares a bracket, J must identically vanish. Thus, the equation $J = 0$ exactly cuts out the 2nd secant variety $\sigma_2(v_4(\mathbb{P}^1))$ in $\mathbb{P}^4$.

- **The Ternary Cubic:** The generic rank is $r = 4$. The Aronhold invariant S has degree $k = 4$ symbols and forms a K_4 tetrahedron graph. If a cubic degenerates to rank $r = 3$, we distribute $k = 4$ symbols among 3 linear forms. A collision is forced. Since every pair of symbols in S shares a bracket, S must vanish. Thus, $S = 0$ exactly cuts out the 3rd secant variety $\sigma_3(v_3(\mathbb{P}^2))$ in $\mathbb{P}^9$.

In this way, the generalisation of the symbolic method to quantic forms provides the explicit algebraic defining equations for the secant geometry of higher-dimensional projective spaces [CCG12]. A purely symbolic, 19th-century hyper graph construction proves to be the exact language required to map the boundaries of modern symmetric tensor rank

5 A Brief Expedition into Geometric Invariant Theory

While the combinatorial machinery of the symbolic method provides a robust algebraic calculus for generating invariants and covariants, its profound utility lies in its geometric interpretation. In the context of projective geometry and Geometric Invariant Theory (GIT), the vanishing of specific symbolic bracket polynomials corresponds directly to distinguished geometric configurations—such as coincident roots on the projective line $\mathbb{P}^1$, or the presence of specific singularities and tangent behaviors on a plane curve in $\mathbb{P}^2$.

Before proceeding to the abstract construction of general moduli spaces, it is essential to ground this algebraic formalism in concrete visual intuition. The following examples translate the abstract determinant brackets and dummy variables into explicit geometric pictures, illustrating precisely how classical algebraic invariants dictate the geometric stability and isomorphism classes of low-degree forms.

5.1 The Quotient $\mathbb{P}(V_d)SL_2$ as a Moduli Space

The previous chapters treated invariants and covariants of binary forms from a classical, computational point of view. We now adopt the modern geometric perspective of Mumford’s Geometric Invariant Theory (GIT) [MFK94]. The foundational philosophy of GIT is that a ring of invariants should be regarded as the coordinate ring of a quotient variety, and that the geometry of this quotient rigorously encodes a moduli problem.

Let G be a complex reductive algebraic group, such as $SL_n(\mathbb{C})$, acting on a projective variety $X \subset \mathbb{P}^N$. The graded ring of invariants is defined as $R(X, G) := \bigoplus_{m \geq 0} \mathbb{C}[X]_m^G$, where $\mathbb{C}[X]_m^G$ denotes the vector space of G -invariant homogeneous polynomials of degree m on X . The *projective GIT quotient* of X by G is then defined as the projective variety $X \backslash G := \text{Proj}(R(X, G))$. In modern algebraic geometry, the Proj construction takes a graded

commutative ring and outputs a projective geometric space where the ring generators act as the homogeneous coordinates. While Mumford's theory guarantees the existence of this quotient space, it is abstract and non-constructive. However, for the space of binary forms $X = \mathbb{P}(V_{2,d})$ under the action of $G = \mathrm{SL}_2(\mathbb{C})$, the First Fundamental Theorem of Invariant Theory dictates that $R(X, G)$ is generated entirely by fully contracted symbolic bracket polynomials [Dol03]. Thus, the symbolic method is the explicit computational engine that builds the coordinate ring of the GIT quotient.

The natural rational map $\pi : X \dashrightarrow X/G$ is not defined everywhere. Its domain is governed by the concept of stability. A point $x \in X$ is called *semistable* if there exists a non-constant homogeneous invariant polynomial that does not vanish at x . In our framework, a binary form is semistable if we can construct at least one symbolic bracket monomial that evaluates to a non-zero value on it. It is *stable* if, additionally, its orbit is closed in the semistable locus and its stabiliser is finite. Points that are not semistable are called *unstable*, forming the *nullcone* of the representation. Algebraically, a binary form lies in the nullcone if and only if *every single fully contracted bracket polynomial evaluates to zero* on it.

To determine stability without checking infinitely many bracket combinations, we rely on the Hilbert–Mumford numerical criterion. This reduces the question to the behavior of the form under 1-parameter subgroups (1-PS), which are homomorphisms of the form $\lambda : \mathbb{C}^* \rightarrow G$.

Theorem 5.1 (Hilbert–Mumford Criterion for Binary Forms). *A binary form $f \in V_{2,d}$ is GIT unstable under the action of $\mathrm{SL}_2(\mathbb{C})$ if and only if it possesses a root of multiplicity strictly greater than $d/2$.*

Proof. By the Hilbert–Mumford numerical criterion, f is unstable if and only if there exists a 1-PS $\lambda : \mathbb{C}^* \rightarrow \mathrm{SL}_2(\mathbb{C})$ such that $\lim_{t \rightarrow 0} \lambda(t) \cdot f = 0$. Up to conjugation, any 1-PS in $\mathrm{SL}_2(\mathbb{C})$ can be diagonalised to $\lambda(t) = \begin{pmatrix} t & 0 \\ 0 & t^{-1} \end{pmatrix}$. Let the binary form be written in the standard binomial basis $f(x, y) = \sum_{i=0}^d \binom{d}{i} a_i x^{d-i} y^i$. The action of $\lambda(t)$ on the coordinates is given by $\lambda(t)^{-1} \cdot x = t^{-1}x$ and $\lambda(t)^{-1} \cdot y = ty$. Therefore, the induced action on the form is:

$$(\lambda(t) \cdot f)(x, y) = f(t^{-1}x, ty) = \sum_{i=0}^d \binom{d}{i} a_i t^{2i-d} x^{d-i} y^i.$$

The coefficient of $x^{d-i} y^i$ under this action is $a_i(t) = a_i t^{2i-d}$. For the limit $\lim_{t \rightarrow 0} \lambda(t) \cdot f$ to exist and equal zero, the exponent of t for every non-zero coefficient must be strictly positive. That is, for all i where $a_i \neq 0$, we must have $2i - d > 0 \implies i > d/2$.

This forces $a_i = 0$ for all $i \leq d/2$. Consequently, the polynomial factors as $f(x, y) = y^{\lfloor d/2 \rfloor + 1} P(x, y)$ for some polynomial P . This means that $y = 0$ (corresponding to the

projective point $[1 : 0]$ is a root of multiplicity at least $\lfloor d/2 \rfloor + 1 > d/2$. Conversely, if f has a root of multiplicity $> d/2$, we can apply an element of $\mathrm{SL}_2(\mathbb{C})$ to move this root to $[1 : 0]$. In these coordinates, $a_i = 0$ for all $i \leq d/2$, and the diagonal 1-PS $\lambda(t)$ will drive the form to 0 as $t \rightarrow 0$. Hence, f is unstable. $\square$

This theorem explicitly links the vanishing of symbolic brackets to root multiplicities, allowing us to interpret the GIT quotient $\mathcal{M}_d^{\mathrm{bin}} := \mathbb{P}(V_{2,d}) \setminus \mathrm{SL}_2$ as a compactified moduli space of unordered point configurations on $\mathbb{P}^1$. For low degrees, this geometry aligns perfectly with our earlier symbolic computations. For a quadratic ($d = 2$), $d/2 = 1$. A form is unstable if it has a double root ($m_{\max} = 2 > 1$). This means the nullcone is exactly the locus where the fundamental symbolic discriminant bracket $\Delta = \frac{1}{2}[\alpha\beta]^2$ evaluates to zero. For a cubic ($d = 3$), a multiple root ($m_{\max} \geq 2$) exceeds $d/2 = 1.5$, making the form unstable. Again, the nullcone is cut out precisely by the vanishing of the massive bracket polynomial $\Delta \propto [\alpha\beta]^2[\gamma\delta]^2[\alpha\gamma][\beta\delta]$.

For quartics ($d = 4$), the threshold is $d/2 = 2$. A quartic is stable if it has four distinct roots. It is *strictly semistable* if it has a double root pattern (e.g., x^2y^2), and unstable if it has a triple root. As established in the previous chapter, stable binary quartics are parametrised by the j -invariant, which is constructed entirely from the symbolic brackets $I = [\alpha\beta]^4$ and $J \propto [\alpha\beta]^2[\beta\gamma]^2[\gamma\alpha]^2$. The GIT quotient $\mathcal{M}_4^{\mathrm{bin}}$ is exactly the projective line $\mathbb{P}^1$, with the rational map $\pi : \mathbb{P}(V_{2,4}) \dashrightarrow \mathbb{P}^1$ given explicitly by the invariant combination $[I^3 : J^2]$. Thus, the abstract Proj construction of Mumford's moduli space is seamlessly and explicitly realised by the algorithms of the symbolic method.

5.2 Plane Curves and their GIT Quotients

We now move from binary forms to plane curves. Let $W \cong \mathbb{C}^3$ and $V_{3,d} = \mathrm{Sym}^d(W^\vee)$ denote the space of homogeneous polynomials of degree d in three variables (x, y, z) . A non-zero element of $V_{3,d}$ defines a plane curve $C \subset \mathbb{P}^2 = \mathbb{P}(W)$ of degree d , well-defined up to scale. Thus, the projective space $\mathbb{P}(V_{3,d}) \cong \mathbb{P}^N$, where $N = \binom{d+2}{2} - 1$, forms the complete linear system of plane curves of degree d .

The special linear group $\mathrm{SL}_3(\mathbb{C})$ acts naturally on $\mathbb{P}^2$ and hence on $\mathbb{P}(V_{3,d})$ by change of coordinates. We consider the projective GIT quotient:

$$\mathcal{M}_d^{\mathrm{plane}} := \mathbb{P}(V_{3,d}) \setminus \mathrm{SL}_3, \quad (30)$$

which is a projective variety parametrising S-equivalence classes of semistable degree- d plane curves. Just as in the binary case, the global coordinates of this moduli space are provided entirely by the symbolic method. The invariant ring of $\mathrm{SL}_3(\mathbb{C})$ acting on ternary forms is generated by fully contracted 3×3 symbolic brackets $[\alpha\beta\gamma]$ [Dol03].

To understand which plane curves are discarded by the GIT quotient, we apply the

Hilbert–Mumford criterion using 1-parameter subgroups $\lambda(t) = \text{diag}(t^{w_1}, t^{w_2}, t^{w_3})$ with $w_1 + w_2 + w_3 = 0$. The action of λ assigns a weight $iw_1 + jw_2 + kw_3$ to every monomial $x^i y^j z^k$. A curve is unstable if all monomials present in its defining equation have strictly positive weights for some choice of w_i .

Geometrically, this algebraic condition translates directly into restrictions on the singularities of the curve. Let $p \in C$ be a singular point of multiplicity m . By placing p at the coordinate origin $[0 : 0 : 1]$ and analysing the resulting Newton polygon of the form, Mumford derived explicit bounds for stability [MFK94]:

- If $m > 2d/3$, the curve is **unstable**.
- If $m = 2d/3$, the curve is at best **strictly semistable** (depending on the configuration of its tangent lines at p).
- If $m < 2d/3$ for all points, and the curve contains no multiple line components, it is generally **stable**.

This multiplicity bound makes the geometry of $\mathcal{M}_d^{\text{plane}}$ remarkably explicit for low degrees. For $d = 3$, the critical multiplicity threshold is $2(3)/3 = 2$. Therefore, any point of multiplicity 3 (a triple point, meaning the curve is a union of three lines intersecting at a single point) is unstable. If a curve has a point of multiplicity $m = 2$, its stability depends on the tangent cone. If the two tangent lines are distinct (an ordinary double point, or *node*), the curve is strictly semistable. If the two tangent lines coincide (a *cusp*), the curve is unstable [Muk03].

The symbolic method makes this abstract classification computationally concrete. As established in the previous chapter, the coordinate ring of the moduli space of cubics is generated by the symbolic hypergraph invariants S (degree 4) and T (degree 6). The GIT quotient is precisely the weighted projective space $\text{Proj}(\mathbb{C}[S, T]) \cong \mathbb{P}(4, 6)$. The unstable nullcone of cuspidal and highly degenerate cubics is defined by the simultaneous vanishing $S = T = 0$. The strictly semistable boundary (nodal cubics) is detected by the vanishing of the discriminant syzygy $\Delta = T^2 - 64S^3 = 0$. For all stable, smooth cubics ($\Delta \neq 0$), the geometric isomorphism class is classified by the absolute j -invariant, constructed algebraically as $j \propto S^3/\Delta$.

For plane quartics ($d = 4$), the quotient $\mathcal{M}_4^{\text{plane}}$ provides a natural compactification for the moduli space of non-hyperelliptic genus 3 curves. The stability threshold rises to $2(4)/3 = 8/3$. Because $2 < 8/3$, curves with ordinary double points *and* curves with cusps are both strictly **stable** when $d = 4$, a striking contrast to the cubic case. A curve only becomes strictly semistable if it possesses a tacnode (two branches sharing a tangent with contact order 2), and becomes unstable if it possesses a triple point ($m = 3 > 8/3$).

For higher degrees d , the geometry of the nullcone and the invariant ring becomes rapidly more complicated. Constructing the entire coordinate ring $\text{Proj}(R(X, G))$ directly via

symbolic brackets becomes computationally prohibitive. Consequently, understanding the birational geometry—and specifically proving the rationality—of $\mathcal{M}_d^{\text{plane}}$ requires a more sophisticated application of the symbolic method: using classical covariants to map between moduli spaces of different degrees.

5.3 Covariants and the Rationality of Moduli Plane Curves

One of the central open problems in algebraic geometry is the *rationality problem*: given a linear representation V of a reductive group G , is the quotient variety $V \backslash G$ birationally equivalent to a projective space? In the context of this chapter, this asks whether the GIT quotient $\mathcal{M}_d^{\text{plane}}$, which compactifies the moduli space of plane curves of degree d , is a rational variety. Extensive work over several decades has shown that $\mathcal{M}_d^{\text{plane}}$ is rational for many low degrees. For large d , however, a powerful modern approach developed by Katsylo, Shepherd-Barron, and later Böhning and Graf von Bothmer, eschews the direct computation of invariants entirely. Instead, it uses classical *covariants* to algebraically map between moduli spaces of different degrees [Dol03].

A covariant of plane curves of degree d is a $\text{PGL}_3(\mathbb{C})$ -equivariant polynomial map $K : V_{3,d} \rightarrow V_{3,d'}$. Concretely, given a homogeneous form $F(x, y, z)$ of degree d , the covariant K assigns another homogeneous form $K(F)$ of degree d' in a way that is perfectly compatible with projective coordinate changes. This algebraic construction induces a rational map of projective spaces $\mathbb{P}(V_{3,d}) \dashrightarrow \mathbb{P}(V_{3,d'})$ given by $[F] \mapsto [K(F)]$. Because the map is equivariant, it descends directly to a rational map between the corresponding GIT quotients, $\mathcal{M}_d^{\text{plane}} \dashrightarrow \mathcal{M}_{d'}^{\text{plane}}$. If we can find a covariant whose generic fiber is a rational variety of known dimension, and if the target moduli space $\mathcal{M}_{d'}^{\text{plane}}$ is already known to be rational, we can deduce the rationality of the original space $\mathcal{M}_d^{\text{plane}}$.

In a major breakthrough, Böhning and Graf von Bothmer successfully applied this strategy by constructing explicit covariants $S_d, T_d : V_{3,d} \rightarrow V_{3,4}$ for plane curves of sufficiently large degree d [BB08; BBK09]. These maps are constructed entirely using the ternary version of the symbolic method. In this formalism, while a pure invariant is formed strictly by multiplying fully contracted 3×3 determinant brackets $[\alpha\beta\gamma]$, a covariant is constructed by multiplying these determinant brackets with symbolic linear factors $\alpha_x = \alpha_1x + \alpha_2y + \alpha_3z$. The brackets ensure the $\text{SL}_3(\mathbb{C})$ -equivariance, while the uncontracted linear factors provide the free variables (x, y, z) necessary to define the new target curve of degree d' . This construction directly generalises the foundational 19th-century algorithms of Clebsch and Gordan into a modern geometric framework.

For a general plane curve C of degree d , the associated pair of quartic curves $(S_d(C), T_d(C))$ is well-defined and varies algebraically with C . The combined map $\Phi_d : \mathbb{P}(V_{3,d}) \dashrightarrow \mathbb{P}(V_{3,4}) \times \mathbb{P}(V_{3,4})$ given by $[F] \mapsto ([S_d(F)], [T_d(F)])$ is $\text{PGL}_3(\mathbb{C})$ -equivariant and dominant onto its image. Passing to the GIT quotients, Φ_d induces a dominant rational map $\bar{\Phi}_d :$

$\mathcal{M}_d^{\text{plane}} \dashrightarrow \mathcal{M}_4^{\text{plane}}$ whose generic fiber is birational to a projective space. Because the moduli space of plane quartics $\mathcal{M}_4^{\text{plane}}$ is classically known to be rational—as it canonically parametrises non-hyperelliptic curves of genus 3—they were able to deduce a profound result:

Theorem 5.2 (Böhring and Graf von Bothmer). *For all sufficiently large degrees d , the moduli space of plane curves $\mathcal{M}_d^{\text{plane}}$ is a rational variety.*

The proof of this modern theorem relies crucially on establishing the non-degeneracy of the covariants S_d and T_d . To prove that these covariant maps do not trivially vanish or collapse the geometry, Böhring and Graf von Bothmer had to evaluate intricate symbolic polynomials. In particular, their key technical lemmas rely on massive symbolic expressions whose vanishing would force a degeneracy contradicting general position. Proving that these expressions are not identically zero is computationally impossible via direct algebraic expansion, as the number of terms would be astronomical.

Instead, they required a highly sophisticated combinatorial analysis of the symbolic bracket monomials. To prove non-vanishing, one evaluates the symbolic expression on a carefully chosen test curve—such as a Fermat-type curve or a highly symmetric Waring decomposition. Just as we evaluated the Aronhold invariants in Chapter ??, the multilinearity of the symbolic brackets $[\alpha\beta\gamma]$, combined with the Pigeonhole principle, forces massive, systematic combinatorial cancellations. By showing that the bracket polynomial simplifies to a single non-zero leading term on this specific geometric locus, they rigorously proved that the covariant is globally non-degenerate.

Thus, the invariant rings and covariants studied by classical algebraists are far more than historical curiosities. The symbolic method—a purely combinatorial algorithm operating on dummy variables, determinant brackets, and linear factors—remains the fundamental, explicitly computable bridge to the deepest rationality problems in modern algebraic geometry.

6 The Noether Code

In 1907, Emmy Noether published her doctoral dissertation, *Über die Bildung des Formensystems der ternären biquadratischen Form* [Noe07], whilst working under the supervision of Gordan. In her paper, Noether applied the symbolic method to the ternary biquadratic form, the plane quartic curve $f = a_x^4$, and successfully computed its complete system of 331 irreducible invariants and covariants. Through manual algebraic calculations, Noether executed the ternary Cayley Ω operations (also known as the "faltung operator", which means folding in German), and applied Plücker straightening algorithms to thousands of bracket polynomials (see [Olv99], or [Stu08] for further details on this algorithm).

However, performing these symbolic manipulations by hand is often awkward. As

demonstrated in the previous chapter, applying classical covariants to modern problems requires computing expressions far too massive for manual derivation. To bridge this gap, the classical algorithms must be digitised.

In this section, we develop a computational framework where we implement the symbolic method developed in Section 2 and the invariant construction algorithm in Section 3.5 is a computational frame. The aim is to translate the algebraic construction of invariants into an algorithmic procedure, enabling systematic generation and verification of invariant polynomials. This is written in Python and uses the SymPy library to translate Noether's algebraic approach to invariant construction into a computational procedure. The full source code is provided in Appendix A.

Here, symbolic brackets are realised as determinants of coefficient matrices, the Cayley Ω -operator is implemented as a differential contraction procedure acting on symbolic products, and bracket monomials correspond to the generated candidate expressions from which invariants are constructed.

6.1 Implementing Symbolic Contraction via the Cayley- Ω Operator

We implement the contraction process underlying the symbolic method via the Cayley Ω -operator. Classically, this process corresponds to collecting bracket factors through alternating differentiation. Computationally, this is a differential contraction procedure acting on products of symbolic forms.

Concretely, the operator acts on products of symbolic expressions of the form

$$(a^{(1)}x^{(1)})^{d_1} \dots (a^{(r)}x^{(r)})^{d_r},$$

The contraction procedure applies alternating differentiation across these variables, systematically extracting bracket factors of the form

$$[a^{(i_1)} \dots a^{(i_n)}],$$

which encode the SL_n -invariant structure of the construction.

The resulting expressions are made up of bracket monomials together with leftover symbolic factors $(a^{(i)}x)^{d_i-r}$. After identifying all symbols $a^{(i)} \sim a$, these expressions evaluate to polynomials in the coefficients of the underlying form. In the contracted case, where there is no dependence on the variables, the process yields invariant polynomials.

6.2 Geometric Evaluation on the Hesse Pencil

Next, we evaluate the invariants on structured families of forms in order to extract some geometric meaning. In particular, the Hesse pencil provides a natural one-parameter family of plane cubics so we can study invariants across the moduli space.

The family of cubic forms,

$$f_\lambda(x, y, z) = x^3 + y^3 + z^3 - 3\lambda xyz,$$

for $\lambda \in \mathbb{C}$, defines the Hesse Pencil. As we have previously computed, this family exhibits rich geometric structure, including singular points corresponding to special values of λ .

Given a candidate invariant produced by the Noether Code, we evaluate it on f_λ by substituting the corresponding coefficients into the polynomial expression obtained after contraction and symbol identification. This yields a function of the parameter λ , whose behaviour reflects geometric properties of the family.

This evaluation serves two purposes. First, it provides a verification of the correctness of the computed invariants, as known invariants illustrate characteristic behaviour on the Hesse pencil. Secondly, it reveals geometric information, such as the detection of singular loci or special configurations, thereby connecting the symbolic construction to geometric invariant theory.

6.3 Searching for Invariants

The final component of the Noether Code is the control loop, implemented in `main.py`, which sets up the generation and verification of new invariants.

The search is carried out over bracket monomials whose symbolic degree matches the invariant degree predicted by the First Fundamental Theorem.

Starting from symbolic expressions of the form

$$(a^{(1)}x^{(1)})^d \dots (a^{(r)}x^{(r)})^d,$$

we construct all bracket monomials by forming determinants

$$[a^{(i_1)} \dots a^{(i_n)}],$$

subject to degree and symmetry constraints compatible with SL_n -invariance. These bracket monomials serve as candidates for invariant expressions.

The Cayley Ω -operator is then applied to perform contraction, producing expressions in bracket monomials and residual symbolic factors. After identifying all symbols $a^{(i)} \sim a$, these expressions are expanded as polynomials in the coefficients of the underlying form, yielding candidate invariants.

With the intention of controlling the combinatorial growth of the search space, we impose restrictions on the total degree and symmetry of the bracket monomials. Nevertheless, the set of candidates grows as degree increases, showing the classical complexity of

generating invariants.

Let us use the ternary quartic to begin with. $f = a_x^4$. Firstly, the code places the polynomial into a breadth-first search (BFS) queue. Then pairs of forms are extracted and the Cayley Ω -operator is applied across all permitted values of k .

When the program discovers a potential new covariant, it is passed to the `groebner_independence_check` module. The algorithm maps the symbolic brackets and linear factors to abstract geometric variables, constructing a polynomial ideal from the basis set using auxiliary variables $I_0, I_1, \dots, I_n$. Using Buchberger's algorithm natively within SymPy, it computes the lexicographic Gröbner basis of this ideal. The code performs a multivariate polynomial reduction of the new candidate covariant modulo this Gröbner basis.

If the remainder from the polynomial division depends only on the auxiliary basis variables I_k , the new covariant is mathematically determined to be a syzygy and is discarded. However, if geometric variables remain in the reduced remainder, the algorithm has proven that the covariant is algebraically independent. The new form is then added to the basis set, as well as the search queue, and the generation loop continues.

6.4 Example Output

The following is the raw output generated by `main.py` when we run the generation loop on the ternary quartic $f = a_x^4$. In this demonstration the engine successfully rejects the vanishing $k = 1$ transvectant, while validating and adding the Hessian ($k = 2$) and the fundamental contravariant ($k = 4$) to the basis set.

```
Fully Autonomous Noether Engine Started
```

```
Starting with basis: [a_x**4]
```

```
Beginning Transvection Loop...
```

```
[Op 2] Found Candidate (k=2): (a b u)**2 * a_x**2 * b_x**2
      [Running Gröbner Independence Check...]
      -> VALIDATED! Added to Basis Set.
```

```
[Op 4] Found Candidate (k=4): (a b u)**4
      [Running Gröbner Independence Check...]
      -> VALIDATED! Added to Basis Set.
```

```
End of Loop
```

```
Total independent generators found: 3
```

```
1. a_x**4
```


2. $(a \ b \ u)^{**2} * a_x^{**2} * b_x^{**2}$
3. $(a \ b \ u)^{**4}$

In summary, the Noether Code is a modern interpretation of Noether’s brute-force calculations. It is a scalable computational tool that can construct the high-degree covariants that may be required for modern moduli space rationality proofs.

7 The Classics Do Indeed Live On

In this dissertation we have followed the evolution of invariant theory from its classical algorithmic foundations to its modern geometric applications. By formalising the classical symbolic method, we demonstrated how variables and determinant brackets encode the continuous symmetries of projective geometry. Applying these classical tools to the framework of Geometric Invariant Theory (GIT) revealed how nineteenth-century covariants remain indispensable in contemporary open questions, specifically the rationality of moduli spaces of plane curves.

Furthermore, we went on to bridge the historical and the modern through computational mathematics. The development of the Noether Code successfully digitised the extensive manual computations of Noether’s 1907 thesis. By mechanising the ternary Faltung operator, evaluating expressions on the Hesse pencil, and using the Gröbner bases for autonomous syzygy detection, the engine transforms the abstract symbolic method into an accessible, explicit computer algorithm, providing a robust framework that generates the high-degree covariants required by modern rationality proofs.

Looking forward, the natural extension of this research lies in scaling the computational engine to investigate plane quintics and higher-degree curves, where the combinatorial complexity currently exceeds known classical bounds. Exploring the connections between these algebraic GIT quotients and their gauge-theoretic counterparts, particularly through the moduli spaces of connections and Yang-Mills theory, has potential to reveal deeper dualities between the combinatorial symbolic method and modern differential geometry.

As this dissertation has shown, the symbolic method is not an artefact from the history of invariant theory. Rather, it is a necessary tool that remains to drive geometric discovery. Thus, the classics do, indeed, live on.

A Source Code: The Noether Code

Code Declaration and AI Assistance

The Python computational engine presented below was developed to algorithmically implement the symbolic method, specifically focusing on the multilinear evaluation of invariants and the computation of transvectants.

While the mathematics, algorithmic logic, and invariant theory applications (such as the evaluation on the Hesse pencil and the extraction of Gröbner bases) are original work, the AI assistant Perplexity (Perplexity AI) was utilised to assist in the programming implementation [Per26]. Specifically, AI assistance was used to:

- Structure the object-oriented Python classes for symbolic bracket representation.
- Optimize the recursive polynomial expansion and term-collection algorithms using the SymPy library.
- Generate standard code for data structures and matrix operations.

All AI-generated code was subsequently reviewed, mathematically verified, debugged, and integrated into the final engine to ensure strict adherence to the classical invariant theory. The complete, verified source code is provided below.

A.1 Module 1: noether_engine.py

This module defines the Bracket class, enforcing the alternating tensor property, and implements the Cayley Ω -process and Plücker straightening algorithms.

```
1 import itertools
2 from sympy import Expr, Symbol, S, sympify, Add, Mul, default_sort_key
3
4 def permutation_parity(lst):
5     inversions = 0
6     for i in range(len(lst)):
7         for j in range(i + 1, len(lst)):
8             if str(lst[i]) > str(lst[j]):
9                 inversions += 1
10    return -1 if inversions % 2 != 0 else 1
11
12 class Linear(Expr):
13     """Represents the linear product a_x"""
14     is_commutative = True
15     def __new__(cls, symbol, variable):
16         return Expr.__new__(cls, sympify(symbol), sympify(variable))
17
18     def _sympystr(self, printer):
19         return f"{self.args[0]}_{self.args[1]}"
```

```

20
21 class Bracket(Expr):
22     """Represents the determinant bracket (a b c)"""
23     is_commutative = True
24     def __new__(cls, *args):
25         args = [sympify(arg) for arg in args]
26
27         # Repeated symbols in a determinant evaluate to 0
28         if len(set(args)) < len(args):
29             return S.Zero
30
31         # Sort alphabetically and calculate parity sign
32         sorted_args = sorted(args, key=str)
33         sign = permutation_parity(args)
34
35         obj = Expr.__new__(cls, *sorted_args)
36         return obj if sign == 1 else -obj
37
38     def _sympystr(self, printer):
39         return f"({' '.join(printer._print(arg) for arg in self.args)})"
40
41 def canonicalize_monomial(term, alphabet):
42     coeff, core_term = term.as_coeff_Mul()
43     if core_term == S.One:
44         return term
45
46     used_dummies = [sym for sym in core_term.free_symbols if sym in alphabet]
47     k = len(used_dummies)
48     if k == 0:
49         return term
50
51     target_dummies = alphabet[:k]
52     mapped_forms = set()
53
54     for p in itertools.permutations(target_dummies):
55         sub_dict = dict(zip(used_dummies, p))
56         new_term = core_term.subs(sub_dict, simultaneous=True)
57         mapped_forms.add(new_term)
58
59     # If swapping dummy variables negates the form, it is 0
60     for form in mapped_forms:
61         if -form in mapped_forms:
62             return S.Zero
63
64     canonical_core = sorted(list(mapped_forms), key=default_sort_key)[0]
65     return coeff * canonical_core
66
67 def canonicalize_expression(expr, alphabet):
68     expr = expr.expand()

```

```

69     terms = Add.make_args(expr)
70     canonical_terms = [canonicalize_monomial(t, alphabet) for t in terms]
71     return Add(*canonical_terms)
72
73 def apply_plucker_to_monomial(term):
74     coeff, core = term.as_coeff_Mul()
75     if core == S.One or not isinstance(core, Mul):
76         return term
77
78     factors = core.args
79     brackets = [f for f in factors if isinstance(f, Bracket)]
80     linears = [f for f in factors if isinstance(f, Linear)]
81
82     if not brackets or not linears:
83         return term
84
85     for bracket in brackets:
86         B_args = list(bracket.args)
87         max_B_sym = B_args[2] # The largest symbol in the sorted bracket
88
89         for linear in linears:
90             l_sym = linear.args[0]
91             var = linear.args[1]
92
93             # If a linear symbol is smaller than the largest bracket symbol,
94             # rewrite!
95             if str(l_sym) < str(max_B_sym):
96                 remaining_factors = list(factors)
97                 remaining_factors.remove(bracket)
98                 remaining_factors.remove(linear)
99                 remainder = Mul(*remaining_factors)
100
101                 p, q, r = B_args[0], B_args[1], B_args[2]
102                 s = l_sym
103
104                 # Identity: (p q r)s_x = (q r s)p_x - (p r s)q_x + (p q s)r_x
105                 term1 = Bracket(q, r, s) * Linear(p, var)
106                 term2 = -Bracket(p, r, s) * Linear(q, var)
107                 term3 = Bracket(p, q, s) * Linear(r, var)
108
109                 new_expression = coeff * remainder * (term1 + term2 + term3)
110
111                 # Recursively apply in case new terms need straightening
112                 return apply_plucker_recursive(new_expression)
113
114     return term
115
116 def apply_plucker_recursive(expr):
117     expr = expr.expand()

```

```

117     if isinstance(expr, Add):
118         return Add(*[apply_plucker_to_monomial(t) for t in expr.args])
119     else:
120         return apply_plucker_to_monomial(expr)
121
122
123 def faltung_ternary(form_F, form_G, k, dummy_F, dummy_G, dual_var):
124     """
125     Computes the k-th ternary transvection of form_F and form_G.
126     Replaces k instances of (dummy_F)_x and (dummy_G)_x with the bracket.
127     """
128     def extract_linears(term, symbol):
129         # 1. If it's a single item (not multiplied by anything)
130         if not isinstance(term, Mul):
131             if isinstance(term, Linear) and term.args[0] == symbol:
132                 return 1, S.One
133             if term.is_Pow and isinstance(term.base, Linear) and term.base.args
134 [0] == symbol:
135                 return term.exp, S.One
136             return 0, term
137
138         linears_count = 0
139         other_factors = []
140         for factor in term.args:
141             if isinstance(factor, Linear) and factor.args[0] == symbol:
142                 linears_count += 1
143             elif factor.is_Pow and isinstance(factor.base, Linear) and factor.
144 base.args[0] == symbol:
145                 linears_count += factor.exp
146             else:
147                 other_factors.append(factor)
148
149         return linears_count, Mul(*other_factors)
150
151     count_F, remainder_F = extract_linears(form_F, dummy_F)
152     count_G, remainder_G = extract_linears(form_G, dummy_G)
153
154     # Cannot transvect if we don't have enough linear factors
155     if count_F < k or count_G < k:
156         return S.Zero
157
158     new_bracket = Bracket(dummy_F, dummy_G, dual_var)
159     rem_linears_F = Linear(dummy_F, 'x')**(count_F - k) if count_F - k > 0 else
160 S.One
161     rem_linears_G = Linear(dummy_G, 'x')**(count_G - k) if count_G - k > 0 else
162 S.One

```

```

160     return remainder_F * remainder_G * rem_linears_F * rem_linears_G * (
        new_bracket**k)

```

Listing 1: noether_engine.py

A.2 Module 2: evaluator.py

This module maps symbolic determinants into explicit geometric parametrisations, identifying the tensor coefficients corresponding to the Hesse pencil.

```

1 from sympy import symbols, expand, Add, S, Poly
2 from noether_engine import Bracket
3
4 def evaluate_on_hesse_pencil(bracket_expr, alphabet, lambda_sym):
5     def expand_bracket(b):
6         a, b_sym, c = b.args
7         a1, a2, a3 = symbols(f'{a}1 {a}2 {a}3')
8         b1, b2, b3 = symbols(f'{b_sym}1 {b_sym}2 {b_sym}3')
9         c1, c2, c3 = symbols(f'{c}1 {c}2 {c}3')
10        return (a1*b2*c3 - a1*b3*c2 - a2*b1*c3 + a2*b3*c1 + a3*b1*c2 - a3*b2*c1
11    )
12
13    #Replace all bracket objects with standard polynomial variables
14    expr_expanded = bracket_expr
15
16    for b in bracket_expr.find(Bracket):
17        expr_expanded = expr_expanded.subs(b, expand_bracket(b))
18
19    # Multiply it out into a standard polynomial
20    expr_expanded = expand(expr_expanded)
21
22    # Collect all the variables
23    all_vars = set()
24    for dummy in alphabet:
25        all_vars.update(symbols(f'{dummy}1 {dummy}2 {dummy}3'))
26
27    # Convert to a formal SymPy Poly object for degree checking
28    poly_expr = Poly(expr_expanded, *all_vars)
29
30    result = S.Zero
31
32    for powers, coeff in poly_expr.terms():
33
34        term_value = coeff
35        valid_term = True
36        var_list = list(poly_expr.gens)
37
38        for dummy in alphabet:
39            v1, v2, v3 = symbols(f'{dummy}1 {dummy}2 {dummy}3')

```

```

40     deg1 = powers[var_list.index(v1)] if v1 in var_list else 0
41     deg2 = powers[var_list.index(v2)] if v2 in var_list else 0
42     deg3 = powers[var_list.index(v3)] if v3 in var_list else 0
43
44     total_deg = deg1 + deg2 + deg3
45     if total_deg == 0:
46         continue
47
48     # Apply Hesse identification rules
49     if deg1 == 3 or deg2 == 3 or deg3 == 3:
50         term_value *= 1
51     elif deg1 == 1 and deg2 == 1 and deg3 == 1:
52         term_value *= lambda_sym
53     else:
54         valid_term = False
55         break
56
57     if valid_term:
58         result += term_value
59
60     return result

```

A.3 Module 3: groebner_checker.py

This module verifies the algebraic independence of the generated covariants by constructing a Gröbner ideal and performing polynomial reduction.

```

1     from sympy import symbols, groebner, reduced
2
3 def groebner_independence_check(new_invariant, basis_set, variables):
4     """
5     Checks if a newly discovered invariant is mathematically independent.
6     If it is a syzygy, it can be written entirely as a polynomial of the
7     basis_set.
8     """
9     if not basis_set:
10         return True
11
12     # Create abstract variables for the invariants: I_0, I_1, I_2...
13     I_vars = symbols(f'I_{0}:{len(basis_set)}')
14
15     # The Ideal is I_k - base_form_k = 0
16     ideal_equations = []
17     for i, base_form in enumerate(basis_set):
18         ideal_equations.append(I_vars[i] - base_form)
19
20     try:
21         # Map our custom Bracket and Linear objects to standard SymPy symbols (
22         Z_0, Z_1...)

```

```

21     # This prevents the polynomial engine from getting confused by class
    attributes.
22     custom_objects = set()
23     for eq in ideal_equations + [new_invariant]:
24         for atom in eq.atoms():
25             if type(atom).__name__ in ['Bracket', 'Linear']:
26                 custom_objects.add(atom)
27         try:
28             from noether_engine import Bracket, Linear
29             custom_objects.update(eq.find(Bracket))
30             custom_objects.update(eq.find(Linear))
31         except:
32             pass
33
34     sub_map = {}
35     for i, obj in enumerate(custom_objects):
36         sub_map[obj] = symbols(f'Z_{i}')
37
38     mapped_ideal = [eq.subs(sub_map) for eq in ideal_equations]
39     mapped_new = new_invariant.subs(sub_map)
40
41     # Z_vars MUST come first so the algorithm eliminates them
42     Z_vars = list(sub_map.values())
43     mapped_vars = Z_vars + list(I_vars)
44
45     # Compute Groebner basis
46     G = groebner(mapped_ideal, *mapped_vars, order='lex')
47
48     # Reduce the new invariant modulo the ideal
49     remainder = reduced(mapped_new, G, *mapped_vars)[1]
50
51     # Check if the reduction successfully eliminated all geometric Z_vars.
52     # If the remainder only contains I_vars, it is a syzygy!
53     remainder_vars = remainder.free_symbols
54
55     if any(z in remainder_vars for z in Z_vars):
56         return True    # It contains geometric variables: it's a NEW
generator!
57     else:
58         return False  # It is a syzygy: purely a polynomial of I_vars
59
60 except Exception as e:
61     print(f" [Groebner Warning] Could not compute basis: {e}")
62     return True

```

A.4 Module 4: main.py

This script runs the breadth-first search (BFS) queue, driving the transvection, straightening, and verification loop.


```

1   from collections import deque
2   from sympy import symbols, S
3   from noether_engine import (
4       Linear, Bracket, canonicalize_expression,
5       apply_plucker_recursive, faltung_ternary
6   )
7   from groebner_checker import groebner_independence_check
8
9   def is_new_generator(new_form, basis_set, all_vars):
10       """
11       Two-step verification for new invariants.
12       1. Fast symbolic core check (catches scalar multiples instantly).
13       2. Groebner syzygy check.
14       """
15       #Fast Core Check
16       new_coeff, new_core = new_form.as_coeff_Mul()
17       for existing_form in basis_set:
18           exist_coeff, exist_core = existing_form.as_coeff_Mul()
19           if new_core == exist_core:
20               return False
21
22       #Groebner Check
23       print("    [Running Gr bner Independence Check...]")
24       return groebner_independence_check(new_form, basis_set, all_vars)
25
26   def main():
27       print("The Noether Code has Started \n")
28
29       #Define alphabet and variables
30       a, b, c, d = symbols('a b c d')
31       alphabet = [a, b, c, d]
32       x, u = symbols('x u')
33       all_vars = [a, b, c, d, x, u]
34
35       #Initialize the basis set with the foundational quartic form
36       f = Linear(a, x)**4
37       basis_set = [f]
38
39       #Setup the BFS Queue
40       queue = deque()
41
42       # Fold f with itself
43       f_a = Linear(a, x)**4
44       f_b = Linear(b, x)**4
45       queue.append((f_a, f_b))
46
47       print(f"Starting with basis: {basis_set}\n")
48       print("Beginning Transvection Loop...\n")
49

```

```

50 operations = 0
51 max_operations = 4 # Running k=1, 2, 3, 4 for the quartic
52
53 while queue and operations < max_operations:
54     form_F, form_G = queue.popleft()
55
56     for k in [1, 2, 3, 4]:
57         operations += 1
58
59         # Transvect
60         raw_form = faltung_ternary(form_F, form_G, k, dummy_F=a, dummy_G=b,
        dual_var=u)
61
62         if raw_form == S.Zero:
63             continue
64
65         # Plücker & Symmetry
66         straightened = apply_plucker_recursive(raw_form)
67         canonical = canonicalize_expression(straightened, alphabet)
68
69         # Syzygy Detection
70         if canonical != S.Zero:
71             print(f"[Op {operations}] Found Candidate (k={k}): {canonical
72             }")
73
74             if is_new_generator(canonical, basis_set, all_vars):
75                 print(f" -> VALIDATED! Added to Basis Set.\n")
76                 basis_set.append(canonical)
77             else:
78                 print(f" -> REJECTED! Syzygy detected.\n")
79
80         print("End of Loop")
81         print(f"Total independent generators found: {len(basis_set)}")
82         for i, gen in enumerate(basis_set):
83             print(f"{i+1}. {gen}")
84
85 if __name__ == "__main__":
86     main()

```

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